Relativistic Dynamics

Conservation laws have to be statements about four-vectors, or they are not laws. Everything else in this chapter is the bill for that sentence.

WHERE WE ARE

Chapters 2.2 to 2.4 rebuilt *kinematics*. We know how coordinates transform, we know what is invariant, we know what a tensor is and why writing a law as an equation between tensors makes it frame-independent by inspection. What we do not yet have is any *mechanics*. Momentum is still $m\mathbf{v}$, energy is still $\frac{1}{2}mv^2$, and force is still $m\mathbf{a}$ — all three inherited unchanged from Chapter 1.1, all three built on a Galilean picture that Chapter 2.2 destroyed.

This chapter rebuilds them. It has exactly one organising idea, and it is worth stating before any algebra so you can watch it do all the work:

Momentum conservation, as you were taught it, is a statement about three numbers. Three numbers are not a frame-independent object. So the law as stated cannot be fundamental.

Repairing it is not optional and the repair is not free. There is essentially one four-vector you can build out of a particle's mass and motion, and once you insist that *it* is the conserved thing, you have also — without asking for it, without any experimental input, and without any way to refuse — acquired a fourth conserved quantity whose low-speed limit is $\frac{1}{2}mv^2$ plus a constant. The constant is mc^2. Everything famous in this chapter is a consequence of that one structural demand, and by §3 you will have watched $E = mc^2$ arrive as an unwanted guest rather than a revelation.

TOOLS YOU'LL NEED — Chapter 0.3 §2: the binomial series $(1+x)^\alpha$ for arbitrary real α, and its worked example 2, which already expanded γmc^2 without saying what it was. Chapter 0.6 §5: the multivariable chain rule, used constantly below to convert $\mathrm{d}/\mathrm{d}\tau$ into $\gamma\,\mathrm{d}/\mathrm{d}t$. Chapter 1.2 §3.5 and §8.1: the Euler–Lagrange equation, and the table of actions this book will write down — §5 here supplies the second entry in that table. Chapter 1.4 §3: Noether's theorem, in particular that invariance under time translation gives a conserved energy; we use it in §5 as an independent check. Chapter 2.3 §5 (proper time as the length of a worldline), §6 (straight worldlines *maximise* it), §4 (the causal classification of four-vectors), and §7 (four-velocity, sketched there and built properly here). Chapter 2.4 §4 (raising and lowering), §5 (the operations preserving tensor character, and the quotient theorem), and above all §6 — the invariance theorem, which is the load-bearing wall of this entire chapter.

1 · Velocity is not a four-vector, and the repair

Start with the object we would most like to keep. A particle traces a worldline through spacetime, and the obvious thing to call its velocity is the rate at which its coordinates change with time:

$$V^\mu \equiv \frac{\mathrm{d}x^\mu}{\mathrm{d}t} = (c,\ \mathbf{v}), \qquad \mathbf{v} = \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t}. \tag{2.5.1}$$

Four numbers, built from a four-vector, containing exactly the information you want. It is also not a four-vector, and the reason is worth stating precisely rather than gesturing at.

1.1 · Exactly what goes wrong

Being a four-vector means one thing and one thing only, by Chapter 2.4 §2: the four components must transform as $A'^\mu = \Lambda^\mu{}_\nu A^\nu$. Test (2.5.1). The numerator is fine: $\mathrm{d}x^\mu$ is the prototype four-vector, so $\mathrm{d}x'^\mu = \Lambda^\mu{}_\nu \mathrm{d}x^\nu$. The denominator is the problem. t is not a scalar; it is the zeroth *component* of a four-vector, and under a boost along x it becomes

$$\mathrm{d}t' = \gamma\left(\mathrm{d}t - \frac{\beta}{c}\,\mathrm{d}x\right) = \gamma\,\mathrm{d}t\left(1 - \frac{\beta v_x}{c}\right), \tag{2.5.2}$$

where the second form divides out $\mathrm{d}t$ and uses $\mathrm{d}x/\mathrm{d}t = v_x$. So

$$V'^\mu = \frac{\mathrm{d}x'^\mu}{\mathrm{d}t'} = \frac{\Lambda^\mu{}_\nu\,\mathrm{d}x^\nu}{\gamma\,\mathrm{d}t\,(1-\beta v_x/c)} = \frac{1}{\gamma\,(1-\beta v_x/c)}\,\Lambda^\mu{}_\nu V^\nu. \tag{2.5.3}$$

Read that carefully. V^μ transforms *almost* correctly — it picks up the right matrix Λ — and then it is spoiled by an extra scalar factor. Worse, that factor depends on the particle's own velocity v_x, so two different particles observed in the same boost get multiplied by two different numbers, and no rescaling of the definition can repair it. This single line is the whole reason Chapter 2.2's velocity-addition law was that awkward quotient rather than a matrix acting on a column: the awkwardness in the denominator of $u' = (u+v)/(1+uv/c^2)$ *is* the factor in (2.5.3).

State the diagnosis in a form that tells you the cure. We divided a four-vector by a frame-dependent quantity. So divide by a frame-*independent* one instead. Chapter 2.3 §5 supplied exactly one natural candidate along a worldline: the proper time τ, which every observer computes and every observer agrees on.

1.2 · The bridge: $\mathrm{d}t/\mathrm{d}\tau = \gamma$

Before we can use τ we need to know how it relates to coordinate time. This is a two-line recomputation of a result from Chapter 2.3 §5, and it is worth having in front of you because we will use it perhaps thirty times. Proper time along a worldline is defined by $c^2\mathrm{d}\tau^2 = \mathrm{d}s^2$, and $\mathrm{d}s^2$ evaluated between two neighbouring events on the worldline is

$$\begin{aligned} \mathrm{d}s^2 &= c^2\mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2 \\ &= c^2\mathrm{d}t^2\left[1 - \frac{1}{c^2}\left(\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2\right)\right]. \end{aligned} \tag{2.5.4}$$

The bracket is $1 - v^2/c^2 = 1/\gamma^2$, so $c^2\mathrm{d}\tau^2 = c^2\mathrm{d}t^2/\gamma^2$ and, taking the positive root (both τ and t increase toward the future),

$$\boxed{\mathrm{d}\tau = \frac{\mathrm{d}t}{\gamma} \qquad \Longleftrightarrow \qquad \frac{\mathrm{d}t}{\mathrm{d}\tau} = \gamma.} \tag{2.5.5}$$

Two remarks that will save confusion later. First, γ here is the particle's own γ, evaluated with its instantaneous speed in whatever frame you are using — it is a function of t and has nothing to do with the γ of any boost between frames. Second, (2.5.5) is the reason $\mathrm{d}/\mathrm{d}\tau$ is computable at all: whenever a τ-derivative appears, you convert it to the t-derivative you know how to take,

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \;=\; \frac{\mathrm{d}t}{\mathrm{d}\tau}\,\frac{\mathrm{d}}{\mathrm{d}t} \;=\; \gamma\,\frac{\mathrm{d}}{\mathrm{d}t}, \tag{2.5.6}$$

which is nothing but the chain rule of Chapter 0.6 §5 applied to a one-parameter reparametrisation.

1.3 · The four-velocity

Now define what we were forced to define:

$$u^\mu \;\equiv\; \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}. \tag{2.5.7}$$

This *is* a four-vector, and the proof is one sentence: the numerator transforms with Λ and the denominator is a scalar, so $u'^\mu = \Lambda^\mu{}_\nu u^\nu$ with no leftover factor. That is the entire content of the definition, and it is the reason relativity keeps replacing $\mathrm{d}/\mathrm{d}t$ by $\mathrm{d}/\mathrm{d}\tau$ everywhere.

Its components come straight from (2.5.6):

$$u^\mu \;=\; \gamma\,\frac{\mathrm{d}x^\mu}{\mathrm{d}t} \;=\; \gamma\,\big(c,\ \mathbf{v}\big) \;=\; \big(\gamma c,\ \gamma v_x,\ \gamma v_y,\ \gamma v_z\big). \tag{2.5.8}$$

Note what happened, because it is the pattern of the whole chapter: **the repaired object is the old object times** γ. At low speed $\gamma \to 1$ and $u^\mu \to (c, \mathbf{v})$, so nothing familiar is lost; at high speed the components run off to infinity, which is exactly the behaviour that lets a fixed-length object point in ever more extreme directions without ever exceeding the speed limit. Which brings us to the property that makes u^μ special.

1.4 · $u \cdot u = c^2$, by explicit contraction

Contract u^μ with itself using the metric, as 2.4 §4.3 instructs — time component keeps its sign, spatial components flip:

$$
\begin{aligned}
u \cdot u \;\equiv\; \eta_{\mu\nu} u^\mu u^\nu \;&=\; \left(u^0\right)^2 - \left(u^1\right)^2 - \left(u^2\right)^2 - \left(u^3\right)^2 \\
&=\; \gamma^2 c^2 - \gamma^2 \left(v_x^2 + v_y^2 + v_z^2\right) \\
&=\; \gamma^2 c^2 \left(1 - \frac{v^2}{c^2}\right) \\
&=\; \gamma^2 c^2 \cdot \frac{1}{\gamma^2} \;=\; c^2.
\end{aligned}
\tag{2.5.9}
$$

Identically. Not for special particles, not in special frames — for every particle, at every instant, in every frame, whatever it is doing. The four-velocity is a four-vector of **fixed Minkowski length** c; all it can do is point in different future-timelike directions.

There is a second, shorter proof that explains *why* rather than merely confirming, and it is the one to remember:

$$
u \cdot u \;=\; \eta_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} \frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} \;=\; \frac{\eta_{\mu\nu}\, \mathrm{d}x^\mu \mathrm{d}x^\nu}{\mathrm{d}\tau^2} \;=\; \frac{\mathrm{d}s^2}{\mathrm{d}\tau^2} \;=\; \frac{c^2 \mathrm{d}\tau^2}{\mathrm{d}\tau^2} \;=\; c^2.
\tag{2.5.10}
$$

The constraint is not a fact about particles. It is the *definition of* τ, rearranged. We chose to parametrise the worldline by its own arc length, and a curve parametrised by arc length has a unit tangent vector; here "unit" means length c because we measured arc length in seconds and tangent vectors in metres per second. Everything in this chapter that looks like a miracle is this sentence in disguise.

1.5 · Four-acceleration, and an orthogonality you get for free

Differentiate again with respect to the invariant:

$$
a^\mu \;\equiv\; \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} \;=\; \frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2}.
\tag{2.5.11}
$$

Same argument as before, twice over: a^μ is a four-vector. Now do something that costs one line and pays for the rest of the chapter — differentiate the identity (2.5.9). Its right-hand side is the constant c^2, so

$$
0 \;=\; \frac{\mathrm{d}}{\mathrm{d}\tau}\left(\eta_{\mu\nu} u^\mu u^\nu\right) \;=\; \eta_{\mu\nu} \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} u^\nu + \eta_{\mu\nu} u^\mu \frac{\mathrm{d}u^\nu}{\mathrm{d}\tau} \;=\; 2\, \eta_{\mu\nu} a^\mu u^\nu,
\tag{2.5.12}
$$

where the two terms merged because $\eta_{\mu\nu}$ is a constant array and is symmetric, so the second sum is the first with the dummy names swapped. Hence

$$
\boxed{u \cdot a \;=\; 0}
\tag{2.5.13}
$$

— always, for every worldline. In Minkowski language, **four-acceleration is orthogonal to four-velocity**. The word "orthogonal" is doing honest work here: it is the same bilinear form, the same definition of perpendicularity, that Chapter 0.5 gave for Euclidean space, with one sign changed.

WHAT THE ORTHOGONALITY MEANS

In Euclidean geometry, if a point moves on a sphere of fixed radius, its velocity is tangent to the sphere — perpendicular to the radius vector. That is the same statement as (2.5.13), one dimension up and one sign over. The four-velocity lives on the hyperboloid $u \cdot u = c^2$; acceleration can only move it *along* that surface, never off it.

So there is no such thing as accelerating "faster through spacetime". A force can turn your four-velocity, tilting it further from your original time axis and thereby increasing γ without limit, but it cannot lengthen it. The speed limit is not a barrier the particle runs into; it is a statement that the only motion available is *rotation of a fixed-length vector*, and a rotation never gets you off the surface you started on. Chapter 2.3 §3.2 drew this surface as the invariant hyperbolae; here it is again, doing dynamics.

It also does immediate accounting work. $u \cdot a = 0$ says the four-acceleration has three independent components, not four — one is always fixed by the other three. In §6 that single constraint turns into the work–energy theorem, which you will therefore not have to assume.

▼ **Grind box — components of a^μ, and $u \cdot a = 0$ checked the hard way**

We will need the components in §6, and computing them is a good drill in (2.5.6). First, the derivative of γ. With $\gamma = (1 - v^2/c^2)^{-1/2}$ and $v^2 = \mathbf{v} \cdot \mathbf{v}$, the chain rule gives

$$\frac{\mathrm{d}\gamma}{\mathrm{d}t} = -\frac{1}{2}\left(1 - \frac{v^2}{c^2}\right)^{-3/2} \cdot \left(-\frac{1}{c^2}\frac{\mathrm{d}(\mathbf{v}\cdot\mathbf{v})}{\mathrm{d}t}\right) = \frac{\gamma^3}{c^2}\,\mathbf{v}\cdot\mathbf{a},$$

using $\mathrm{d}(\mathbf{v}\cdot\mathbf{v})/\mathrm{d}t = 2\mathbf{v}\cdot\mathbf{a}$ with $\mathbf{a} \equiv \mathrm{d}\mathbf{v}/\mathrm{d}t$ the ordinary three-acceleration. Write $\dot\gamma$ for this. Now apply (2.5.6) to (2.5.8):

$$a^\mu = \gamma\frac{\mathrm{d}}{\mathrm{d}t}(\gamma c,\ \gamma\mathbf{v}) = \Big(\gamma\dot\gamma c,\ \ \gamma\dot\gamma\,\mathbf{v} + \gamma^2\mathbf{a}\Big).$$

Now contract, carefully, keeping every term:

$$\begin{aligned} u\cdot a &= (\gamma c)(\gamma\dot\gamma c) - (\gamma\mathbf{v})\cdot(\gamma\dot\gamma\mathbf{v} + \gamma^2\mathbf{a}) \\ &= \gamma^2\dot\gamma c^2 - \gamma^2\dot\gamma v^2 - \gamma^3\,\mathbf{v}\cdot\mathbf{a} \\ &= \gamma^2\dot\gamma c^2\left(1 - \frac{v^2}{c^2}\right) - \gamma^3\,\mathbf{v}\cdot\mathbf{a} \\ &= \dot\gamma c^2 - \gamma^3\,\mathbf{v}\cdot\mathbf{a}. \end{aligned}$$

Substituting $\dot\gamma = \gamma^3(\mathbf{v}\cdot\mathbf{a})/c^2$ from the first line makes the two terms cancel identically. ✓ The abstract one-liner (2.5.12) and the brute-force component calculation agree, as they must; the point of doing both once is to see how much labour the index notation is saving.

One consequence worth extracting now. Go to the particle's instantaneous rest frame, where $\mathbf{v} = 0$ and $u^\mu = (c, \mathbf{0})$. Then $u \cdot a = c\, a^0 = 0$ forces $a^0 = 0$, so in that frame $a^\mu = (0, \mathbf{a}_0)$ where $\mathbf{a}_0$ is the acceleration the particle itself feels — call it the **proper acceleration**. Its invariant square is

$$a \cdot a = 0 - |\mathbf{a}_0|^2 = -\,\alpha^2, \qquad \alpha \equiv |\mathbf{a}_0|\,,$$

so **four-acceleration is always a spacelike vector**, and its invariant magnitude is the proper acceleration — what an accelerometer bolted to the particle reads. That number is frame-independent, which is why "the rocket burns at $1g$" is a meaningful statement while "the rocket's acceleration is $9.8\,\mathrm{m\,s^{-2}}$" is a statement about somebody's coordinates. Problem 2 uses this.

2 · Four-momentum, and the argument this chapter turns on

We have a four-vector that describes a particle's motion and a scalar that describes the particle itself — its rest mass m, which by the conventions of this Part *always* means the rest mass and never anything else. Multiplying a four-vector by a scalar gives a four-vector (2.4 §5.1, operation 2), so define the **four-momentum**

$$p^\mu \;\equiv\; m\,u^\mu \;=\; \gamma m\,\big(c,\;\mathbf{v}\big) \;=\; \big(\gamma mc,\;\;\gamma m\mathbf{v}\big). \tag{2.5.14}$$

Its spatial part is $\gamma m\mathbf{v}$, which reduces to the Newtonian $m\mathbf{v}$ when $\gamma \to 1$, so we have at least a candidate for "momentum". Call it $\mathbf{p} \equiv \gamma m\mathbf{v}$. The time component $p^0 = \gamma mc$ is, at this stage, an unnamed number that came along for the ride. We are not going to name it until §3, and we are not going to name it by guessing — the name will be forced.

Everything now rests on a single question: *why should this be the conserved quantity?* The answer is the intellectual centre of the chapter, so we take it slowly.

2.1 · What a conservation law has to be

A conservation law is a claim about a physical process: some quantity computed before equals the same quantity computed after. Consider a collision — particles come in, particles come out, and we assert

$$\sum_{\text{in}} Q_i \;=\; \sum_{\text{out}} Q_f. \tag{2.5.15}$$

Now ask the question that Chapter 2.4 taught us to ask about every equation: *who is asserting this, and does the assertion survive translation into another observer's language?* Because if it does not, then (2.5.15) is not a law of nature. It is a report from one laboratory, of no more fundamental standing than "the train is moving".

Chapter 2.4 §6 gave the tool. If the two sides of an equation are tensors of the same type, the equation holds in every frame if it holds in one. If they are not, all bets are off — and Chapter 2.4's closing warning was blunt about which side of that line statements like $\mathbf{p} = 0$ fall on: they are statements about components, hence about a frame, and must never be carried across a boost unexamined.

So the demand is clear. Whatever is conserved must be a **tensor**, and the conservation statement must be an equation between tensors of the same type. Three numbers are not a tensor under the Lorentz group. Four numbers that transform with Λ are.

2.2 · Newtonian momentum conservation does not survive a boost

That is an argument from principle. Here is the failure, in arithmetic, in a collision you can hold in your head.

Two identical lumps of putty, each of rest mass m, approach each other along the x-axis in frame S with speeds $+u$ and $-u$. They collide and stick. By symmetry the composite is at rest in S; call its rest mass M, whatever that turns out to be.

Check Newtonian momentum conservation in S:

$$\underbrace{mu + m(-u)}_{\text{before}} \;=\; 0 \;=\; \underbrace{M \cdot 0}_{\text{after}}. \qquad \checkmark \tag{2.5.16}$$

Perfect. Note how little that check tested: by symmetry it would have passed for *any* definition of momentum of the form $\mathbf{p} = mf(v)\,\mathbf{v}$ with f any function whatever of the speed — the two terms cancel no matter what f is. A symmetric collision in a single frame is almost completely uninformative. The information is in the boost.

So boost. Move to a frame S' travelling at $-v$ along x relative to S, so that in S' everything is drifting in the $+x$ direction. Chapter 2.2's velocity-addition law gives the new velocities of the two lumps:

$$u_1' \;=\; \frac{u+v}{1+uv/c^2}, \qquad\qquad u_2' \;=\; \frac{-u+v}{1-uv/c^2}, \tag{2.5.17}$$

and the composite, at rest in S, moves at v in S'. Now evaluate the Newtonian momentum in S'. Write $a \equiv uv/c^2$ for compactness and put the two fractions over a common denominator $1 - a^2$:

$$\begin{aligned} u_1' + u_2' &= \frac{(u+v)(1-a) + (v-u)(1+a)}{(1+a)(1-a)} \\ &= \frac{\big(u+v-au-av\big) + \big(v-u+av-au\big)}{1-a^2} \\ &= \frac{2v-2au}{1-a^2} \;=\; \frac{2v\left(1-u^2/c^2\right)}{1-u^2v^2/c^4}, \end{aligned} \tag{2.5.18}$$

where the last step used $au = u^2v/c^2$ so that $2v - 2au = 2v(1-u^2/c^2)$. Meanwhile the Newtonian momentum *after* the collision is Mv, and Newtonian mechanics also insists that mass is additive, $M = 2m$. So the two sides of the conservation law read, in S',

$$2mv\underbrace{\frac{1-u^2/c^2}{1-u^2v^2/c^4}}_{\text{before}} \qquad \text{versus} \qquad \underbrace{2mv}_{\text{after}}. \tag{2.5.19}$$

These are equal only if $u = 0$ or $v = 0$ — that is, only if there was no collision or no boost. For $u = v = 0.6c$: the fraction is $(1 - 0.36)/(1 - 0.1296) = 0.64/0.8704 = 0.7353$, so an observer in S' finds the outgoing momentum **36% larger** than the incoming momentum. The books balance in S and fail badly in S'.

> ⚠ **WHAT EXACTLY FAILED**
>
> It is tempting to say "Newtonian momentum isn't conserved at high speed". That is not what the calculation showed. In frame S it was conserved, exactly, to all orders — see (2.5.16). What failed is *agreement between observers about whether it was conserved*. One frame says the law holds; another says it is violated by 36%.
>
> That is a much worse disease than being approximately wrong, and it cannot be cured by a small correction term. A quantity whose conservation is frame-dependent is not describing anything about the collision; it is describing the laboratory. This is precisely the defect Chapter 2.4 §6.2 diagnosed in $\mathbf{F} = m\mathbf{a}$ — "its content changes when you change frames, which disqualifies it as a statement about nature rather than about a laboratory" — now shown rather than asserted.

2.3 · The four-vector version cannot fail

Now run the same collision with p^μ and watch the disease vanish for structural reasons.

Suppose that in frame S the total four-momentum is conserved: $\sum_{\text{in}} p^\mu = \sum_{\text{out}} p^\mu$, all four components. Define the shortfall

$$\Delta P^\mu \equiv \sum_{\text{out}} p^\mu - \sum_{\text{in}} p^\mu. \tag{2.5.20}$$

Each p^μ is a four-vector; sums and differences of tensors of the same type are tensors of that type (2.4 §5.1, operation 1); so ΔP^μ is a four-vector. Our hypothesis is the tensor equation $\Delta P^\mu = 0$, and 2.4 §6 says a tensor equation true in one frame is true in all. Explicitly:

$$\Delta P'^\mu = \Lambda^\mu{}_\nu \, \Delta P^\nu = \Lambda^\mu{}_\nu \cdot 0 = 0. \tag{2.5.21}$$

That is the whole proof, and it is deliberately anticlimactic. The transformation law is linear and homogeneous, so it maps zero to zero; there is nowhere for a violation to come from. **If four-momentum is conserved for one observer, it is conserved for every observer, automatically, with no further conditions.**

▼ **Grind box — the same collision, done relativistically, term by term**

The abstract argument is airtight, but you should see the numbers come out once. We need one identity, and it is worth having permanently.

The velocity-addition identity for γ. If $u' = (u+v)/(1+uv/c^2)$, then

$$1 - \frac{u'^2}{c^2} = \frac{\left(1 + \frac{uv}{c^2}\right)^2 - \frac{(u+v)^2}{c^2}}{\left(1 + \frac{uv}{c^2}\right)^2}.$$

Expand the numerator, being careful with every term:

$$\begin{aligned} \text{num} &= 1 + \frac{2uv}{c^2} + \frac{u^2v^2}{c^4} - \frac{u^2}{c^2} - \frac{2uv}{c^2} - \frac{v^2}{c^2} \\ &= 1 - \frac{u^2}{c^2} - \frac{v^2}{c^2} + \frac{u^2v^2}{c^4} \;=\; \left(1 - \frac{u^2}{c^2}\right)\left(1 - \frac{v^2}{c^2}\right). \end{aligned}$$

The $2uv/c^2$ terms cancel and what remains factorises. Taking reciprocals and square roots,

$$\boxed{\gamma(u') = \gamma(u)\,\gamma(v)\left(1 + \frac{uv}{c^2}\right).}$$

This is just the statement that rapidities add, $\cosh(\phi_u + \phi_v) = \cosh\phi_u \cosh\phi_v + \sinh\phi_u \sinh\phi_v$, written in velocities; Chapter 2.2 §5 proved the rapidity version.

Momentum before, in S'. Using the identity with $u \to u$ and then $u \to -u$:

$$\gamma(u_1')\,u_1' = \gamma(u)\gamma(v)\left(1 + \frac{uv}{c^2}\right)\cdot\frac{u+v}{1+uv/c^2} = \gamma(u)\gamma(v)\,(u+v),$$

$$\gamma(u_2')\,u_2' = \gamma(u)\gamma(v)\left(1 - \frac{uv}{c^2}\right)\cdot\frac{v-u}{1-uv/c^2} = \gamma(u)\gamma(v)\,(v-u).$$

The messy denominators cancel exactly — that is the identity earning its keep. Adding and multiplying by m:

$$\sum_{\text{before}} \gamma m\, u' \;=\; m\gamma(u)\gamma(v)\big[(u+v) + (v-u)\big] \;=\; 2m\,\gamma(u)\gamma(v)\,v.$$

Momentum after, in S'. The composite has rest mass M and moves at v, so its momentum is $\gamma(v)Mv$. Equality demands

$$\gamma(v)\,M\,v = 2m\,\gamma(u)\gamma(v)\,v \qquad \Longrightarrow \qquad M = 2\gamma(u)\,m.$$

Read that twice. Relativistic momentum conservation in the boosted frame does not merely *permit* the composite's rest mass to exceed $2m$ — it **requires** it, and fixes the excess exactly. The lump of putty is heavier than its parts, by a factor $\gamma(u)$, and nothing in the calculation gave us a choice. Section 9 is about that number.

The time components, for free. Do the same with $p^0 = \gamma mc$:

$$\sum_{\text{before}} \gamma mc = mc\,\gamma(u)\gamma(v)\left[\left(1+\frac{uv}{c^2}\right)+\left(1-\frac{uv}{c^2}\right)\right] = 2mc\,\gamma(u)\gamma(v),$$

while after the collision $p^0 = \gamma(v)Mc = 2\gamma(u)\gamma(v)mc$ using the M just derived. Equal ✓ — and note that we did not impose this. It came out. That is (2.5.21) being true in coordinates.

And in S itself. There, before: $p^0 = 2\gamma(u)mc$; after: $p^0 = Mc = 2\gamma(u)mc$ ✓, while the spatial parts vanish on both sides by symmetry. So the four-vector statement holds in S, holds in S', and holds — by (2.5.21) — in every frame there is.

2.4 · The part you cannot refuse: momentum forces energy

So far we have shown that four-momentum conservation is *self-consistent* across frames. Now comes the sharper statement, and it is the one that makes this chapter inevitable. Suppose you are a minimalist. You accept that the conserved object must be built from the four-vector p^μ — you have seen §2.2 and you are not going back — but you want to conserve only the three spatial components, since those are the ones you have experimental evidence for. You want the law "$\Delta \mathbf{P} = 0$ in every frame", and you decline to say anything about ΔP^0.

You cannot have it. Here is why, in three lines.

THEOREM — CONSERVING THREE COMPONENTS FORCES THE FOURTH

Let ΔP^μ be a four-vector. If its three spatial components vanish in *every* inertial frame, then its time component vanishes too.

Proof. By hypothesis $\Delta P^\mu = (\Delta P^0,\ 0,\ 0,\ 0)$ in the original frame. Boost along x with speed βc. By Chapter 2.4's transformation law (2.5.21) the new first spatial component is

$$\Delta P'^1 \;=\; \gamma\big(\Delta P^1 - \beta\,\Delta P^0\big) \;=\; -\gamma\beta\,\Delta P^0.$$

The hypothesis says this must vanish for every β. Since $\gamma\beta \neq 0$ for $\beta \neq 0$, we need $\Delta P^0 = 0$. ■

Look at what that argument did. It used no dynamics, no experiment, no assumption about forces or collisions. It used only the transformation law. And it says: **a universe in which momentum is conserved for every observer is a universe in which p^0 is conserved too.** You are not permitted to conserve three components of a four-vector; the boost mixes the fourth into the first three and drags it into the law whether you invited it or not.

That is the whole chapter in one paragraph. The demand that conservation laws be frame-independent *selects* p^μ out of all the candidate momenta, and having selected it you are stuck with its time component — a fourth conserved quantity you did not ask for, did not derive from any experiment, and cannot discard. Section 3 asks what on earth it is.

⚠ **TWO HONEST CAVEATS ABOUT WHAT HAS BEEN PROVED**

This is not a proof that anything is conserved. Nothing so far says nature conserves p^μ; that is a physical claim, and Chapter 1.4 gave its real origin — Noether's theorem, applied to the invariance of the laws under translations in space (momentum) and time (energy). What we have proved is *conditional*: *if* some four-vector quantity is conserved in one frame, it is conserved in all; and *if* a frame-independent momentum conservation law exists at all, the conserved object must be a four-vector and its time component comes along. Relativity narrows the field of candidates to essentially one. Experiment then confirms that this one is right, and it does so every day in every particle detector on Earth.

The choice of m as the scalar was not forced by tensor algebra alone. $p^\mu = mu^\mu$ is a four-vector, but so is $f(m)u^\mu$ for any function f, and so is $m\,g(m/M_0)\,u^\mu$ for any dimensionless combination you can invent. What fixes the choice is the correspondence limit: as $\gamma \to 1$ the spatial part must become the Newtonian $m\mathbf{v}$ that three centuries of experiment supports. That pins the scalar to m itself. Covariance narrows; correspondence chooses. That pair of moves is the method of the entire second half of this book.

3 · What the time component is

We have an unnamed conserved quantity $c\,p^0 = \gamma mc^2$, with dimensions of energy — check: mass times velocity squared. Dimensions are suggestive and prove nothing. Let us find out what it actually is by the only honest route available: expand it at low speed and see what it turns into.

3.1 · The expansion

Apply the binomial series of Chapter 0.3 §2, which for arbitrary real exponent α reads $(1+x)^\alpha = \sum_{k\geq 0} \binom{\alpha}{k} x^k$ with $\binom{\alpha}{k} = \alpha(\alpha-1)\cdots(\alpha-k+1)/k!$. Take $\alpha = -\frac{1}{2}$ and $x = -\beta^2$, where $\beta = v/c$:

$$\binom{-1/2}{1} = -\frac{1}{2}, \qquad \binom{-1/2}{2} = \frac{(-\frac{1}{2})(-\frac{3}{2})}{2} = \frac{3}{8}, \qquad \binom{-1/2}{3} = \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{6} = -\frac{5}{16}. \tag{2.5.22}$$

Substituting $x = -\beta^2$ flips the sign of every odd power, so all the signs become positive:

$$\gamma \;=\; \left(1-\beta^2\right)^{-1/2} \;=\; 1 + \frac{1}{2}\beta^2 + \frac{3}{8}\beta^4 + \frac{5}{16}\beta^6 + \cdots \tag{2.5.23}$$

Multiply by mc^2 and restore $\beta = v/c$:

$$\gamma mc^2 = \underbrace{mc^2}_{\text{constant}} + \underbrace{\tfrac{1}{2}mv^2}_{\text{Newtonian}} + \underbrace{\frac{3}{8}\frac{mv^4}{c^2}}_{\text{first correction}} + \frac{5}{16}\frac{mv^6}{c^4} + \cdots \tag{2.5.24}$$

3.2 · The identification, made carefully

Now the part that deserves care, because it is usually skated over. What (2.5.24) shows is a mathematical fact about a function, not a physical identification. It says: *the conserved quantity γmc^2 equals a constant, plus the Newtonian kinetic energy, plus terms that vanish as $v/c \to 0$*. Getting from there to "γmc^2 is the energy" requires one further argument, and here it is.

Newtonian mechanics has its own conserved quantity for elastic collisions, $\sum_i \frac{1}{2}m_i v_i^2$, verified to exhaustion at low speeds. Our new law says $\sum_i \gamma_i m_i c^2$ is conserved. Expand the new claim, particle by particle, at low speed:

$$\sum_i \gamma_i m_i c^2 = \underbrace{\sum_i m_i c^2}_{(*)} + \sum_i \tfrac{1}{2}m_i v_i^2 + O\!\left(\frac{v^4}{c^2}\right). \tag{2.5.25}$$

If the collision does not change what the particles *are* — same species in, same species out, which is the only kind of collision Newtonian mechanics ever contemplated — then the sum $(*)$ is the same before and after, and cancels out of the conservation statement entirely. What is left is exactly Newtonian kinetic-energy conservation. So the new law *contains* the old one, and the quantity it conserves must be what the old theory called energy. That, and only that, is what licenses the name:

$$\boxed{E \equiv \gamma mc^2, \qquad p^\mu = \left(\frac{E}{c}, \mathbf{p}\right), \qquad \mathbf{p} = \gamma m\mathbf{v}.} \tag{2.5.26}$$

The four-momentum is the energy and the momentum, welded into one object by the boost that mixes them. From here on p^μ is called the energy–momentum four-vector, and the two conservation laws you learned separately are one law with four components.

THE CONSTANT THAT REFUSES TO BE A CONSTANT

In Newtonian mechanics the zero of energy is arbitrary: adding a constant to E changes no equation of motion, no conservation law, no prediction. That freedom is why nobody ever asked what the "true" energy of a stationary object was — the question had no content.

Relativity removes the freedom, and the mechanism is visible in (2.5.25). The would-be constant is $\sum_i m_i c^2$, and it cancels *only if the rest masses do not change*. The moment a process alters them — a nucleus binds, a particle decays, two lumps of putty stick together and (as §2.3's grind box proved) end up with rest mass $2\gamma m$ rather than $2m$ — the sum $(*)$ is different before and after, and the difference has to be paid for out of kinetic energy. So mc^2 is not an additive constant one may drop. It is a **reservoir**, and processes exist that draw on it.

Setting $v = 0$ in (2.5.26) names the reservoir:

$$E_0 \;=\; mc^2.$$

The **rest energy**. Note what it is not: it is not a formula for the energy of a moving body, which is γmc^2; and it is not a claim that mass "is" energy in some substantial sense. It is the value of the conserved quantity E for a particle that happens to be at rest, and its only physical content is the one just derived — that changes in rest mass show up as changes in kinetic energy, at an exchange rate of c^2.

3.3 · The first correction, and where you can see it

The $\frac{3}{8}mv^4/c^2$ term in (2.5.24) is not decoration. Compare it with the Newtonian term:

$$\frac{\frac{3}{8}mv^4/c^2}{\frac{1}{2}mv^2} \;=\; \frac{3}{4}\frac{v^2}{c^2} \;=\; \frac{3}{4}\beta^2. \tag{2.5.27}$$

Quadratically small — which is why Newtonian mechanics survived so long, and also why the correction becomes visible the moment you have a system with a characteristic speed and a spectroscope. The electron in a hydrogen atom has $\beta \approx \alpha = 1/137$ (Chapter 0.3 §5 got that from dimensional analysis alone: the natural speed $k_e/\hbar$ divided by c is exactly α). So

$$\frac{3}{4}\beta^2 \;\approx\; \frac{3}{4}\alpha^2 \;=\; \frac{3}{4} \times 5.325 \times 10^{-5} \;=\; 3.99 \times 10^{-5}, \tag{2.5.28}$$

a relative shift of four parts in 10^5 in the atom's energy levels. On a $13.6\ \mathrm{eV}$ scale that is of order $\alpha^2 \times 13.6\ \mathrm{eV} \approx 7 \times 10^{-4}\ \mathrm{eV}$, comfortably resolvable, and it is one of the three contributions to the **fine structure** of hydrogen that Chapter 0.3's worked example 2 flagged and could not yet explain. Chapter 4.5 computes the full splitting; the point here is that the leading piece of it is a term in (2.5.24) and nothing more exotic.

▼ **Grind box — the same correction in momentum, and why the coefficient changes sign**

Open a quantum mechanics text and you will find the relativistic correction to the hydrogen Hamiltonian written as $-\hat{p}^4/8m^3c^2$: a *negative* term with coefficient $\frac{1}{8}$. We just derived a *positive* term with coefficient $\frac{3}{8}$. Both are correct. Chapter 0.3's worked example 2 noticed the clash and named the culprit; now that we have actually derived $\mathbf{p} = \gamma m \mathbf{v}$ and the mass shell, we can close it arithmetically.

The kinetic energy in terms of p**.** Anticipating §4's mass shell, $E = \sqrt{p^2c^2 + m^2c^4}$. Factor out mc^2 and expand with the binomial series again, now with $x = p^2/m^2c^2$ and $\alpha = +\frac{1}{2}$, for which $\binom{1/2}{1} = \frac{1}{2}$ and $\binom{1/2}{2} = \frac{(\frac{1}{2})(-\frac{1}{2})}{2} = -\frac{1}{8}$:

$$E = mc^2\left(1 + \frac{p^2}{m^2c^2}\right)^{1/2} = mc^2\left[1 + \frac{p^2}{2m^2c^2} - \frac{p^4}{8m^4c^4} + \cdots\right] = mc^2 + \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} + \cdots$$

There it is, sign and coefficient. Quantum mechanics uses this form because $\hat{p}$ is the natural operator, not v.

Why the two agree. The variable is different: $p = \gamma m v$, not mv. Expand γ to the order needed,

$$p^2 = \gamma^2 m^2 v^2 = m^2v^2\left(1 - \frac{v^2}{c^2}\right)^{-1} = m^2v^2\left(1 + \frac{v^2}{c^2} + \cdots\right),$$

so the two terms become

$$\frac{p^2}{2m} = \frac{mv^2}{2} + \frac{mv^4}{2c^2} + \cdots, \qquad -\frac{p^4}{8m^3c^2} = -\frac{m^4v^4}{8m^3c^2} + \cdots = -\frac{mv^4}{8c^2} + \cdots$$

Add them: $\frac{1}{2}mv^2 + \left(\frac{1}{2} - \frac{1}{8}\right)mv^4/c^2 = \frac{1}{2}mv^2 + \frac{3}{8}mv^4/c^2$ ✓, which is (2.5.24). The apparent discrepancy was bookkeeping: $\frac{1}{2}$ of the quartic term comes from p not being mv, and $-\frac{1}{8}$ from the square root, and $\frac{1}{2} - \frac{1}{8} = \frac{3}{8}$.

The moral is one you will need again in Part V: *a perturbative coefficient is meaningless until you say which variable is being held fixed.* Terms move between orders when you change variables, and only the physical prediction — here, the size of the fine-structure shift, which is $\sim \alpha^2$ either way — is invariant.

4 · The mass shell

Every four-vector has an invariant square, and the invariant square of p^μ turns out to contain everything.

4.1 · $p \cdot p = m^2c^2$, and what it says

Contract $p^\mu = mu^\mu$ with itself. The mass is a scalar, so it comes straight out of the contraction, and (2.5.9) finishes the job:

$$p \cdot p \;=\; \eta_{\mu\nu}\,(mu^\mu)(mu^\nu) \;=\; m^2\,\big(u \cdot u\big) \;=\; m^2c^2. \tag{2.5.29}$$

Now compute the same quantity from the components in (2.5.26), remembering that lowering flips the sign of the spatial parts:

$$p \cdot p \;=\; \big(p^0\big)^2 - |\mathbf{p}|^2 \;=\; \frac{E^2}{c^2} - p^2, \qquad p \equiv |\mathbf{p}|\,. \tag{2.5.30}$$

Set the two expressions equal and multiply through by c^2:

$$\boxed{E^2 \;=\; p^2c^2 + m^2c^4.} \tag{2.5.31}$$

This is the **mass-shell relation**, and it is the most useful equation in particle physics. Three observations before we use it.

It is an invariant statement. The left side involves E, the right involves p; both change under a boost, and the particular combination $E^2 - p^2c^2$ does not. Different observers assign different energies and momenta to the same electron and all compute the same m. So *mass is a Lorentz invariant* — not a quantity that grows with speed, but the fixed label attached to the four-vector, exactly as the interval is the fixed label attached to a pair of events.

It is Chapter 2.3's geometry, in different variables. Plot E against p and (2.5.31) is a hyperbola with asymptote $E = pc$. Compare 2.3 §3.2, where the orbits of a boost in the (ct, x) plane were the hyperbolae $c^2t^2 - x^2 = \text{const}$ with asymptotes at $45°$. Same curves, same reason: a boost is a hyperbolic rotation, and it slides any four-vector along the level surface of its own invariant square. Momentum space is spacetime with the axes relabelled.

It has no reference to τ. That matters more than it looks, and §4.3 collects the debt.

4.2 · Velocity from the four-momentum

Before the massless case — and it must be before, because it is what makes the massless case work — extract the velocity. Both components of p^μ carry the same factor γm, so dividing kills it:

$$\frac{\mathbf{p}}{E} \;=\; \frac{\gamma m\mathbf{v}}{\gamma mc^2} \;=\; \frac{\mathbf{v}}{c^2} \qquad \Longrightarrow \qquad \boxed{\mathbf{v} \;=\; \frac{\mathbf{p}\,c^2}{E}.} \tag{2.5.32}$$

Read it as a recipe: *given a particle's energy and momentum, its velocity is the ratio.* No γ, no m, no square roots. Two checks. At low speed, $E \approx mc^2$ and $\mathbf{p} \approx m\mathbf{v}$, so the ratio returns $\mathbf{v}$ ✓. And in general, since $E \geq pc$ by (2.5.31) whenever $m^2 \geq 0$, the formula automatically gives $v \leq c$ — the speed limit is not imposed on (2.5.32), it falls out of it.

4.3 · Massless particles

Set $m = 0$ in (2.5.31). Taking the positive root, since energies of real particles are positive:

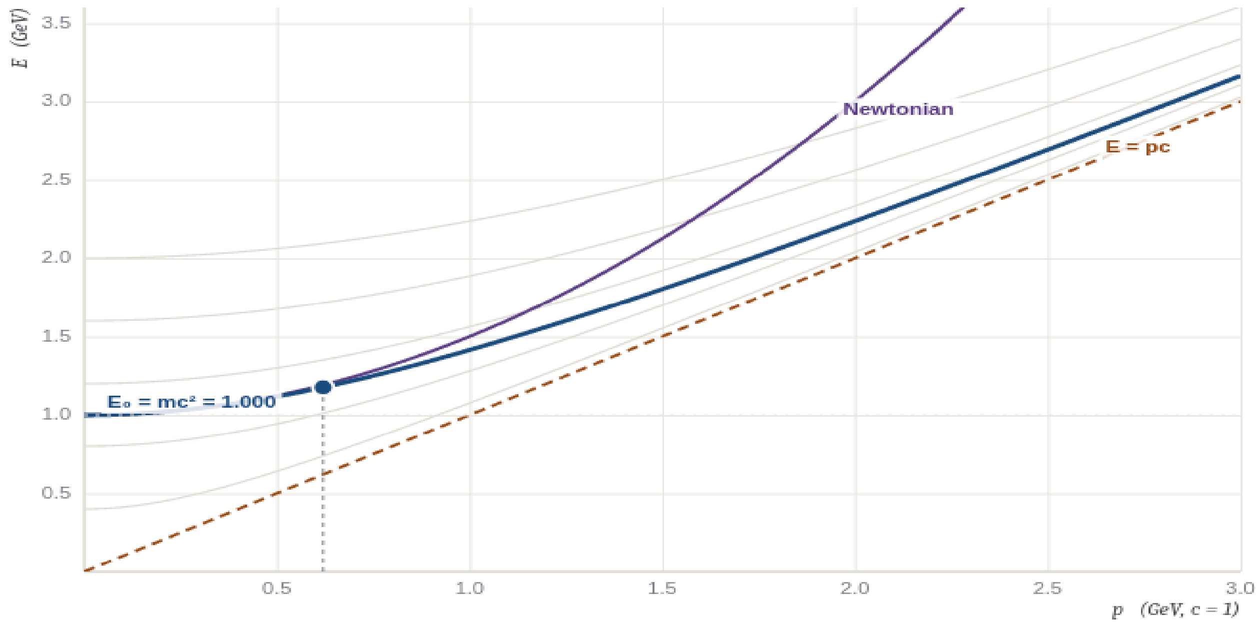
$$E = pc. \tag{2.5.33}$$

Feed that into (2.5.32):

$$v = \frac{pc^2}{E} = \frac{pc^2}{pc} = c. \tag{2.5.34}$$

Exactly c, for every energy, with no limiting process and no approximation. A massless particle does not travel *near* the speed of light; it travels at it, always, and it has no rest frame in which to be at rest.

> ⚠ **WHY THIS ISN'T OBVIOUS — "MASSLESS" IS A STATEMENT ABOUT A FOUR-VECTOR**
>
> It is natural to read $m = 0$ as the end of a sequence: particles get lighter and lighter, and in the limit you have a photon. That reading is wrong, and it is wrong in a way that will cost you.
>
> **The construction $p^\mu = mu^\mu$ does not survive the limit.** As $m \to 0$ at fixed energy, $\gamma \to \infty$, and $p^\mu = mu^\mu$ is $0 \times \infty$ — an indeterminate form, not a limit. Worse, $u^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau$ needs τ, and a null worldline has $\mathrm{d}s^2 = 0$ at every point, so $\mathrm{d}\tau = 0$ identically and there is no proper time to divide by. Photons do not have worldlines parametrised by proper time; a photon has no clock, and "the time experienced by a photon" is not a small number, it is a meaningless phrase.
>
> **The correct definition takes p^μ as primary.** A massless particle is one whose four-momentum is a *null* vector, $p \cdot p = 0$. That is a perfectly good statement with no τ, no γ, and no m multiplying anything. Everything else follows: (2.5.31) gives $E = pc$, (2.5.32) gives $v = c$, and Chapter 2.3 §4's causal classification already told you where such a vector points — along the light cone. Nothing new had to be assumed; the massless case is the null case of a classification we already had.
>
> This inversion — *the four-momentum is fundamental, and mass is the label $p \cdot p$ carries* — is the version that survives into quantum field theory, where particles are excitations of fields and m appears as a parameter in a Lagrangian long before anything is moving.



rest mass *m* momentum *p*

hide Newtonian curve | let m → 0 | a proton | find the 1% point

```
m = 1.000 p = 0.620 E = √(p²+m²) = 1.1766

v/c = pc²/E = 0.5269 γ = E/mc² = 1.1766

Newtonian m + p²/2m = 1.1922 error = 1.325 %

Newtonian error passes 1% at p = 0.571 (v/c = 0.496, γ = 1.152)
```

The mass shell. *Inside this figure only*, units are chosen with $c = 1$ and everything measured in GeV, so E, pc and mc^2 are all just numbers on the same scale and the asymptote $E = pc$ is the line at $45°$. Every curve and every readout is computed numerically from (2.5.31), (2.5.32) and $E_{\text{Newt}} = mc^2 + p^2/2m$; nothing is drawn from a stored result. **(i)** The bold blue curve is the exact hyperbola $E = \sqrt{p^2c^2 + m^2c^4}$; the faint curves behind it are the same relation at other masses — *the invariant hyperbolae of Chapter 2.3 §3.2, now in momentum space*, one for each value of $p \cdot p$. **(ii)** The orange dashed line is the asymptote $E = pc$, the degenerate member $m = 0$ of that family — the light cone of momentum space. Drag the mass slider down and watch the hyperbola flatten onto it; at $m = 0$ the curve *is* the line, and the readout shows v/c locked at exactly 1. **(iii)** The purple curve is the Newtonian energy $mc^2 + p^2/2m$. It leaves the hyperbola with the same value and the same slope at $p = 0$ — the two agree to second order, which is (2.5.24) — and then diverges upward without limit, because Newtonian mechanics believes any energy buys any speed. Move the momentum slider and watch the error readout: it stays under 1% until the particle is moving at half the speed of light, then climbs through 6% at $v = 0.71c$ and 34% at $v = 0.89c$. That quartic-then-catastrophic behaviour is why nobody noticed relativity until they built accelerators. **(iv)** The marker sits on the exact curve at the momentum you choose; drag it, or click anywhere in the frame. Its readouts are E, $\gamma = E/mc^2$, and $v/c = pc/E$ — the last computed from (2.5.32) and not from any γ.

▼ **Grind box — why the Newtonian curve is so good for so long**

The figure claims the Newtonian energy is within 1% of the truth up to $v \approx 0.5c$, which is surprising if you expect errors of order β^2. Here is the reason, and it is a nice piece of Chapter 0.3 bookkeeping.

Work in units of mc and mc^2: write $x = p/mc$, so the exact energy is $E/mc^2 = \sqrt{1+x^2}$ and the Newtonian one is $1 + x^2/2$. The fractional error is

$$\varepsilon(x) \;=\; \frac{1+x^2/2}{\sqrt{1+x^2}} - 1.$$

Expand with the binomial series, $(1+x^2)^{-1/2} = 1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 - \cdots$:

$$\left(1+\frac{x^2}{2}\right)\left(1-\frac{x^2}{2}+\frac{3x^4}{8}\right) = 1 - \frac{x^2}{2} + \frac{3x^4}{8} + \frac{x^2}{2} - \frac{x^4}{4} + O(x^6) = 1 + \frac{x^4}{8} + O(x^6).$$

The x^2 terms cancel *exactly*. That cancellation is not luck: the Newtonian formula was constructed to reproduce the first two terms of (2.5.24), so the leading error is necessarily the third. Hence $\varepsilon \approx x^4/8$ — **quartic**, not quadratic, and a quartic is very small for a while and then is not.

Numbers. Setting $\varepsilon = 0.01$ exactly (numerically, not from the expansion) gives $x = 0.5715$, whence $v/c = x/\sqrt{1+x^2} = 0.4962$ and $\gamma = 1.152$. Some more, exact:

p/mc	v/c	E/mc^2	NEWTONIAN	ERROR
0.5	0.447	1.1180	1.1250	0.62%
1	0.707	1.4142	1.5000	6.07%
2	0.894	2.2361	3.0000	34.2%
3	0.949	3.1623	5.5000	73.9%
5	0.981	5.0990	13.500	165%

This is the quantitative answer to "why didn't anyone notice?". Everything in nineteenth-century mechanics ran at $\beta < 10^{-4}$, where $\varepsilon < 10^{-17}$. You do not find relativity by doing careful mechanics; you find it by doing electromagnetism, which is Chapter 2.1's story, or by building something that goes fast.

⚠ WHY THIS ISN'T OBVIOUS — NOTHING HERE FORBIDS $V > C$ BY DECREE

Look back over §§1–4 and notice what is *not* in them: no axiom saying "speeds cannot exceed c". The prohibition is a consequence, and it arrives twice, by different routes.

Causally (Chapter 2.3 §4.5). A signal travelling faster than c can be boosted into a frame where it arrives before it left, and combining two such signals builds a closed causal loop. That argument came from the geometry alone and applies to influences, not just to objects.

Energetically (this chapter). From (2.5.26), pushing a massive particle to $v \to c$ requires $\gamma \to \infty$ and hence $E \to \infty$. The barrier is not a wall; it is a bill. And (2.5.32) shows the same thing from the other side: no finite E and p satisfying (2.5.31) with $m > 0$ can give $pc^2/E = c$.

The two arguments are independent and they agree, which is the sort of over-determination that makes a structure believable. Note also what neither of them forbids: a hypothetical particle with $p \cdot p < 0$ — spacelike four-momentum, "imaginary mass" — would satisfy $v > c$ happily. Nothing in the algebra excludes it; what excludes it is causality, plus the fact that in quantum field theory such a field signals an unstable vacuum rather than a fast particle. Chapter 6.6 makes that precise, because the Higgs field before symmetry breaking is exactly such a case, and the resolution is not that anything travels faster than light.

5 · The action is proper time

We have the repaired momentum $\mathbf{p} = \gamma m\mathbf{v}$, and we expect the equation of motion to read $\mathrm{d}\mathbf{p}/\mathrm{d}t = \mathbf{F}$ — Newton's second law in the form he actually wrote it, with the repaired momentum substituted for the old one. (Section 6 shows this is the spatial part of a proper tensor equation; take it as the target for now.) Chapter 1.2 established that respectable dynamics comes from an action principle. So: *what Lagrangian produces this?*

We do not guess. We impose the answer and solve for L — which is the honest way round, and is exactly how Chapter 1.2 §4 checked that $L = T - V$ reproduces Newton, run backwards.

5.1 · Solving for L

Take a particle in a potential, $L = L_0(\mathbf{v}) - V(\mathbf{x})$, with the free part depending only on the velocity. The Euler–Lagrange equation for the coordinate x^i — one equation per degree of freedom, $\frac{\mathrm{d}}{\mathrm{d}t}\left(\partial L/\partial \dot{q}^i\right) = \partial L/\partial q^i$, from Chapter 1.2 §3.5 — reads

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L_0}{\partial v^i}\right) = \frac{\partial L}{\partial x^i} = -\frac{\partial V}{\partial x^i} = F_i. \tag{2.5.35}$$

We want the left-hand side to be $\mathrm{d}p_i/\mathrm{d}t$ with $p_i = \gamma m v^i$. Matching the two demands

$$\frac{\partial L_0}{\partial v^i} = \gamma m\, v^i = \frac{m\, v^i}{\sqrt{1 - v^2/c^2}}. \tag{2.5.36}$$

Now solve this for L_0. Since the right-hand side is v^i times a function of the speed alone, L_0 can depend on $\mathbf{v}$ only through $v = |\mathbf{v}|$, and then the chain rule gives $\partial L_0/\partial v^i = L_0'(v)\, \partial v/\partial v^i = L_0'(v)\, v^i/v$. Comparing with (2.5.36), the direction v^i cancels and we are left with a single ordinary differential equation:

$$L_0'(v) = \frac{m\, v}{\sqrt{1 - v^2/c^2}}. \tag{2.5.37}$$

Integrate. Substitute $s = 1 - v^2/c^2$, so $\mathrm{d}s = -2v\,\mathrm{d}v/c^2$ and $v\,\mathrm{d}v = -\frac{c^2}{2}\mathrm{d}s$:

$$\begin{aligned} L_0 &= \int \frac{m\, v\, \mathrm{d}v}{\sqrt{s}} = -\frac{mc^2}{2} \int s^{-1/2}\, \mathrm{d}s \\ &= -\frac{mc^2}{2} \cdot 2\sqrt{s} + \text{const} = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} + \text{const}. \end{aligned} \tag{2.5.38}$$

The constant is genuinely free — Chapter 1.2 §7 showed that adding a constant to L adds a term linear in t to S, which is a total derivative and changes no equation of motion. Set it to zero, for a reason that will be apparent in ten lines:

$$\boxed{L = -mc^2\sqrt{1 - \beta^2} - V(\mathbf{x}) = -\frac{mc^2}{\gamma} - V.} \tag{2.5.39}$$

Check the limit before going on. Expanding the square root with the binomial series, $\sqrt{1 - \beta^2} = 1 - \frac{1}{2}\beta^2 - \frac{1}{8}\beta^4 - \cdots$, so

$$L = -mc^2 + \tfrac{1}{2}mv^2 + \frac{mv^4}{8c^2} + \cdots - V, \tag{2.5.40}$$

which is Chapter 1.2's $T - V$ plus the irrelevant constant $-mc^2$ plus small corrections ✓. Note in passing that L is emphatically *not* $T - V$ any more: $-mc^2/\gamma$ is not the kinetic energy of anything. Chapter 1.2 §5 warned that "$T - V$" is a special case rather than a definition; here is the first place the warning cashes out.

5.2 · The rewrite that makes it obvious

Now form the action and use (2.5.5) in the direction we have not yet used it. For a free particle ($V = 0$),

$$S = \int_{t_1}^{t_2} L\, \mathrm{d}t = -mc^2 \int_{t_1}^{t_2} \sqrt{1 - \beta^2}\, \mathrm{d}t = \boxed{-mc^2 \int \mathrm{d}\tau} = -mc^2\, \tau[\text{path}]. \tag{2.5.41}$$

Stop here. This is the most important equation in the chapter after (2.5.31), and it deserves to be read as a sentence rather than a formula.

THE FREE RELATIVISTIC ACTION IS THE PROPER TIME

Up to a constant $-mc^2$, **the action of a free particle is the reading of the clock it carries.** Not a quantity resembling it, not a quantity proportional to it in some limit — it is that number.

So the two principles this book has stated separately are one principle. Chapter 1.2 said: nature makes S stationary. Chapter 2.3 §6 proved: among all worldlines joining two events, the straight one has the *greatest* proper time. Put them side by side and the minus sign does its job — *maximising* τ is *minimising* $-mc^2\tau$, because $mc^2 > 0$. The minus sign in front of mc^2 is not a convention chosen for tidiness; it is what converts 2.3's maximum into 1.2's stationary point, and Chapter 1.2's grind box on the second variation already checked the Legendre condition for exactly this Lagrangian and found $\partial^2 L/\partial \dot{x}^2 = m\gamma^3 > 0$, confirming that the stationary point is a genuine minimum of S.

Two loops close at once. First, the free particle's trajectory is *the longest worldline*, which is why the travelling twin comes back younger: she took a shorter path in the only sense of "length" spacetime recognises. Second, this is entry two in Chapter 1.2 §8.1's table of actions — quoted forward there, derived here. Whatever it is worth, the derivation cost eleven lines.

And the forward view. Chapter 3.3 keeps (2.5.41) *verbatim* and changes only how $\mathrm{d}\tau$ is computed, replacing $\eta_{\mu\nu}$ by a position-dependent metric $g_{\mu\nu}(x)$. Extremising the same functional then gives the geodesic equation, and general relativity's statement that free particles fall along geodesics is *this* statement, with a different ruler. Nothing about the principle changes at all.

▼ **Grind box — three independent checks on L, including Noether's**

Check 1: the canonical momentum. Chapter 1.2 §6 defines the momentum conjugate to x^i as $\partial L/\partial \dot{x}^i$. From (2.5.39), with $\dot{x}^i = v^i$,

$$\frac{\partial L}{\partial \dot{x}^i} = -mc^2 \cdot \frac{1}{2}\left(1-\beta^2\right)^{-1/2} \cdot \left(-\frac{2v^i}{c^2}\right) = \frac{mv^i}{\sqrt{1-\beta^2}} = \gamma m v^i = p_i. \checkmark$$

Which it had better, since (2.5.36) is what we solved. But it is reassuring that the canonical momentum of Lagrangian mechanics and the spatial part of the four-momentum are literally the same object — they did not have to be.

Check 2: Noether's energy. Chapter 1.4 §3 showed that if L has no explicit time dependence, the conserved Noether charge is $H = \sum_i \dot{x}^i\,\partial L/\partial \dot{x}^i - L$. Compute it for the free case:

$$\begin{aligned} H &= \mathbf{v}\cdot(\gamma m\mathbf{v}) + mc^2\sqrt{1-\beta^2} \\ &= \gamma m v^2 + \frac{mc^2}{\gamma} \\ &= \gamma m\left(v^2 + \frac{c^2}{\gamma^2}\right) \;=\; \gamma m\left(v^2 + c^2 - v^2\right) \;=\; \gamma m c^2. \end{aligned}$$

where the third line used $c^2/\gamma^2 = c^2(1-\beta^2) = c^2 - v^2$. So the Noether charge of time-translation invariance is exactly $E = \gamma mc^2$, the quantity §3 identified by an entirely different argument (matching a Taylor expansion to Newtonian kinetic energy). Two independent derivations, one answer. ✓

Check 3: the equation of motion, in full. Confirm that (2.5.39) really does give the promised law and not something that merely resembles it. With $V = 0$, Euler–Lagrange says $\mathrm{d}(\gamma m v^i)/\mathrm{d}t = 0$, so $\gamma m\mathbf{v}$ is constant, so $\mathbf{v}$ is constant (because γ is a monotonic function of v, so fixing γmv fixes v), so the path is a straight line traversed uniformly. Free particles move in straight lines at constant speed — Newton's first law, recovered from an action whose Lagrangian is nothing like $\frac{1}{2}mv^2$. ✓

An aside on manifest covariance. (2.5.41) is manifestly a scalar — m is a scalar, c is a scalar, τ is a scalar — so the *action* is frame-independent by inspection, in the sense of Chapter 2.4 §6. The *Lagrangian* (2.5.39) is not, and neither is $\mathrm{d}t$; only their product is. This is the standard situation in relativistic mechanics and it is why the action formulation is preferred: the object with the good transformation properties is the integral, not the integrand-in-t.

6 · Force, and the death of relativistic mass

Newton's second law is the one piece of the old mechanics we have not yet replaced. Chapter 2.4 §6.2 already announced the verdict — $\mathbf{F} = m\mathbf{a}$ is not a tensor equation and $\mathrm{d}p^\mu/\mathrm{d}\tau = f^\mu$ is — but announcing is not deriving. Here is the object and here is what it costs.

6.1 · The covariant force

Define the **four-force** by the only construction available: differentiate the four-momentum by the invariant.

$$f^\mu \equiv \frac{\mathrm{d}p^\mu}{\mathrm{d}\tau} = m\,\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = m\,a^\mu, \tag{2.5.42}$$

the last step holding for particles of constant rest mass. Both sides are $(1,0)$ tensors, so by Chapter 2.4 §6 this is a legitimate law: written once, it holds in every frame. And it inherits a constraint at no cost, because $u \cdot a = 0$ from (2.5.13):

$$f \cdot u = m\,(a \cdot u) = 0. \tag{2.5.43}$$

So a four-force has only three independent components: whatever three you specify, the fourth is determined. That is not a defect. It is a theorem with a name you already know, as we are about to see.

6.2 · The three-force, and the work–energy theorem for free

Connect to the laboratory. Define the ordinary **three-force** as the rate of change of the ordinary momentum with respect to ordinary time, $\mathbf{F} \equiv \mathrm{d}\mathbf{p}/\mathrm{d}t$ — which is Newton's second law in its correct form, the form he actually wrote. Using (2.5.6) on (2.5.42), component by component:

$$f^\mu \;=\; \gamma\,\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{E}{c},\,\mathbf{p}\right) \;=\; \gamma\left(\frac{1}{c}\frac{\mathrm{d}E}{\mathrm{d}t},\;\; \mathbf{F}\right). \tag{2.5.44}$$

Now impose (2.5.43) using $u^\mu = \gamma(c, \mathbf{v})$:

$$0 \;=\; f\cdot u \;=\; \gamma\left(\frac{1}{c}\frac{\mathrm{d}E}{\mathrm{d}t}\right)(\gamma c) - \gamma\mathbf{F}\cdot\gamma\mathbf{v} \;=\; \gamma^2\left(\frac{\mathrm{d}E}{\mathrm{d}t} - \mathbf{F}\cdot\mathbf{v}\right), \tag{2.5.45}$$

and since $\gamma^2 \neq 0$,

$$\boxed{\frac{\mathrm{d}E}{\mathrm{d}t} \;=\; \mathbf{F}\cdot\mathbf{v}.} \tag{2.5.46}$$

That is the **work–energy theorem** — the rate of change of energy is the power delivered by the force — and we did not assume it, define it, or import it. It is the time component of a four-vector equation, forced by the geometric identity $u \cdot a = 0$. In Newtonian mechanics the work–energy theorem is a separate derivation; here it is the statement that the four-force is orthogonal to the four-velocity, which is the statement that the four-velocity has fixed length, which is the definition of proper time. Everything is the same thing.

6.3 · $\mathbf{F}$ and $\mathbf{a}$ are not parallel

Now the damage. Expand $\mathbf{F} = \mathrm{d}(\gamma m\mathbf{v})/\mathrm{d}t$ with the product rule, using $\dot\gamma = \gamma^3(\mathbf{v}\cdot\mathbf{a})/c^2$ from §1.5's grind box:

$$\mathbf{F} \;=\; \frac{\mathrm{d}}{\mathrm{d}t}(\gamma m\mathbf{v}) \;=\; \gamma m\,\mathbf{a} + \dot\gamma\, m\mathbf{v} \;=\; \gamma m\left[\mathbf{a} + \frac{\gamma^2}{c^2}(\mathbf{v}\cdot\mathbf{a})\,\mathbf{v}\right]. \tag{2.5.47}$$

The bracket contains a piece along $\mathbf{a}$ and a piece along $\mathbf{v}$. Unless those two directions coincide, or unless the second term vanishes, **the force and the acceleration point in different directions.** Push a fast particle sideways and it does not accelerate sideways; it accelerates partly forward, or partly backward, depending on the sign of $\mathbf{v}\cdot\mathbf{a}$. Nothing in Newtonian mechanics prepares you for this, and it is not a small effect.

The two special cases where they *are* parallel are worth having explicitly.

Longitudinal — push along the motion. Then $\mathbf{a} \parallel \mathbf{v}$, so $\mathbf{v}\cdot\mathbf{a} = va$ and the bracket is $a\big(1+\gamma^2\beta^2\big)\,\hat v$. Use the identity $\gamma^2 - \gamma^2\beta^2 = \gamma^2(1-\beta^2) = 1$, i.e. $1 + \gamma^2\beta^2 = \gamma^2$:

$$F_\parallel \;=\; \gamma m\, a_\parallel\cdot\gamma^2 \;=\; \gamma^3 m\, a_\parallel. \tag{2.5.48}$$

Transverse — push across the motion. Then $\mathbf{v} \cdot \mathbf{a} = 0$, the second term dies, and

$$F_\perp \;=\; \gamma m\, a_\perp. \tag{2.5.49}$$

Two different laws, differing by γ^2, for the same particle at the same instant. At $\gamma = 10$ a longitudinal push is a hundred times less effective at producing acceleration than a transverse one of the same magnitude. This is not an exotic regime: it is routine at any electron accelerator, and it is why the transverse focusing of a beam and its longitudinal acceleration are engineered as separate problems with separate hardware.

▼ **Grind box — inverting the relation: $\mathbf{a}$ in terms of $\mathbf{F}$**

(2.5.47) gives $\mathbf{F}$ from $\mathbf{a}$. For solving actual problems you want the other direction, and inverting it is a small exercise in projection that is worth doing once because the same trick appears in Chapter 3.3.

Step 1 — get the component along $\mathbf{v}$. Dot (2.5.47) with $\mathbf{v}$:

$$\mathbf{F} \cdot \mathbf{v} = \gamma m \left[\mathbf{v} \cdot \mathbf{a} + \frac{\gamma^2}{c^2}(\mathbf{v} \cdot \mathbf{a})v^2\right] = \gamma m\,(\mathbf{v} \cdot \mathbf{a})\,\left[1 + \gamma^2\beta^2\right] = \gamma^3 m\,(\mathbf{v} \cdot \mathbf{a}),$$

using $1 + \gamma^2\beta^2 = \gamma^2$ again. So

$$\mathbf{v} \cdot \mathbf{a} = \frac{\mathbf{F} \cdot \mathbf{v}}{\gamma^3 m}.$$

Step 2 — substitute back. Put that into (2.5.47) and solve for $\mathbf{a}$:

$$\mathbf{F} = \gamma m\,\mathbf{a} + \gamma m \cdot \frac{\gamma^2}{c^2} \cdot \frac{\mathbf{F} \cdot \mathbf{v}}{\gamma^3 m}\,\mathbf{v} = \gamma m\,\mathbf{a} + \frac{(\mathbf{F} \cdot \mathbf{v})}{c^2}\,\mathbf{v},$$

hence

$$\boxed{\mathbf{a} = \frac{1}{\gamma m}\left[\mathbf{F} - \frac{(\mathbf{F} \cdot \mathbf{v})}{c^2}\,\mathbf{v}\right].}$$

Check it. For $\mathbf{F} \parallel \mathbf{v}$: the bracket is $\mathbf{F}(1 - \beta^2) = \mathbf{F}/\gamma^2$, so $a = F/\gamma^3 m$ ✓ matching (2.5.48). For $\mathbf{F} \perp \mathbf{v}$: the second term vanishes, $a = F/\gamma m$ ✓ matching (2.5.49).

What the formula says. The bracket removes from $\mathbf{F}$ the piece $(\mathbf{F} \cdot \hat{v})\beta^2\hat{v}$ — a projection along the motion, weighted by β^2. So the acceleration is a *distorted* image of the force, squashed along the direction of travel. At $\beta \to 1$ the longitudinal response is suppressed entirely: you can push a nearly-light-speed particle as hard as you like along its motion and it barely speeds up, though its energy rises steadily at the rate (2.5.46) says it must. The energy has to go somewhere, and it goes into γ rather than into v.

One consequence worth naming. Set $\mathbf{F}$ constant, along x, starting from rest. Then $\mathrm{d}(\gamma m v)/\mathrm{d}t = F$ integrates immediately to $\gamma m v = Ft$, so

$$v(t) = \frac{Ft/m}{\sqrt{1+(Ft/mc)^2}} \;\longrightarrow\; c \quad \text{as } t \to \infty,$$

approaching but never reaching c, whereas the Newtonian $v = Ft/m$ sails past it at $t = mc/F$. Same constant force, same duration, entirely different destination — and the relativistic answer required no new physical assumption beyond $\mathbf{p} = \gamma m \mathbf{v}$.

6.4 · Why this book will not say "relativistic mass"

There is a tempting way to make (2.5.48) and (2.5.49) look like $\mathbf{F} = m\mathbf{a}$: absorb the γ's into the mass. Define $m_{\text{rel}} = \gamma m$ and you can write $\mathbf{p} = m_{\text{rel}}\mathbf{v}$ and $E = m_{\text{rel}}c^2$, and the formulas look Newtonian again. Textbooks did this for sixty years. It is a mistake, and the calculation above is the cleanest way to see why.

It would have to be two different numbers at once. To keep $\mathbf{F} = m\mathbf{a}$ you need $m_{\parallel} = \gamma^3 m$ for a longitudinal push and $m_{\perp} = \gamma m$ for a transverse one. A "mass" that depends on which way you shove is not a property of the particle; it is a badly chosen name for the components of a tensor relation. And for a general push, by (2.5.47), no scalar whatsoever works, because $\mathbf{F}$ and $\mathbf{a}$ are not even parallel — you would need a matrix, and calling a matrix "the mass" abandons the only thing the word was supposed to mean.

It makes $E = mc^2$ empty. With $m_{\text{rel}} \equiv E/c^2$, the celebrated equation $E = m_{\text{rel}}c^2$ says $E = E$. All the physical content — that a particle at rest still has energy mc^2, that binding changes rest mass, that mass is not additive — lives in the *rest* mass and is invisible in the relativistic one.

It hides the invariant. The real structure is (2.5.31): m is what you get by contracting p^μ with itself, a Lorentz scalar, the same for all observers, in exactly the way the interval is the same for all observers. Redefining "mass" to mean E/c^2 replaces an invariant with a component and throws away the geometry. It is the momentum-space version of saying that a metre stick "really" gets shorter when it moves.

So: m means rest mass, always, everywhere in this book, exactly as the conventions for Part II state. Energy is $E = \gamma mc^2$ and momentum is $\mathbf{p} = \gamma m\mathbf{v}$, and the γ's stay where they are, visible, attached to the motion rather than smuggled into the particle. The phrase "relativistic mass" does not appear again.

7 · Light: the wave four-vector, and Doppler

Everything so far has been about particles. Waves need one more four-vector, and it comes from an argument so simple it is easy to undervalue.

7.1 · Why k^μ has to be a four-vector

A plane wave has a phase

$$\Phi(t, \mathbf{x}) \;=\; \mathbf{k}\cdot\mathbf{x} - \omega t, \tag{2.5.50}$$

with $\mathbf{k}$ the wave vector ($|\mathbf{k}| = 2\pi/\lambda$, pointing along propagation) and ω the angular frequency. Here is the physical claim, and it is the whole argument:

> **COUNTING CRESTS IS FRAME-INDEPENDENT**
>
> A crest is not a thing that moves; it is a set of events at which the field is momentarily maximal. So "is *this event* a crest?" is a yes-or-no question about a single point of spacetime, and both observers are being asked about the same point. They must give the same answer.
>
> Now count. Put a detector at a fixed place and let it click once per crest between two events A and B on its own worldline. The number of clicks is an **integer**. An observer flying past at $0.9c$ watches the same detector and counts the same integer — each click is an event on one worldline, and 2.3 §4.3 proved that the ordering of events on a single timelike worldline is invariant. Nobody can gain or lose a click by changing frames.
>
> That count is $\Delta\Phi/2\pi$. An integer cannot transform, and A and B were arbitrary, so Φ is a **Lorentz scalar**.

Now use the scalar to manufacture the four-vector. Write (2.5.50) in the form of a contraction. With $x^\mu = (ct, x, y, z)$ and the metric lowering the spatial indices,

$$\begin{gathered} \text{define } \; k^\mu \equiv \left(\frac{\omega}{c},\, \mathbf{k}\right); \qquad \text{then} \\ k_\mu x^\mu = k^0(ct) - \mathbf{k}\cdot\mathbf{x} \;=\; \omega t - \mathbf{k}\cdot\mathbf{x} \;=\; -\Phi. \end{gathered} \tag{2.5.51}$$

So $\Phi = -k_\mu x^\mu$. We know x^μ is a four-vector and we have just argued Φ is a scalar — for *every* event x^μ, not merely for one. That is precisely the hypothesis of the quotient theorem (Chapter 2.4 §5.1's grind box): if $k_\mu A^\mu$ is a scalar for every four-vector A^μ, then k_μ is a covector, and raising with η makes k^μ a four-vector. Hence

$$\boxed{k^\mu \;=\; \left(\frac{\omega}{c},\, \mathbf{k}\right) \quad \text{is a four-vector.}} \tag{2.5.52}$$

For light in vacuum the dispersion relation is $\omega = c\,|\mathbf{k}|$ (Chapter 2.1 §3 derived it from Maxwell), which says precisely $k\cdot k = \omega^2/c^2 - |\mathbf{k}|^2 = 0$: **the wave four-vector of light is null**. Set that beside $p\cdot p = 0$ from §4.3 and the shape of Chapter 4.1's punchline is already visible; we come back to it below.

7.2 · Longitudinal Doppler

Everything about the Doppler effect is now one matrix multiplication. Let a light wave travel in the $+x$ direction in frame S, so

$$k^\mu \;=\; \frac{\omega}{c}(1,\ 1,\ 0,\ 0), \tag{2.5.53}$$

the spatial part having magnitude ω/c as the null condition requires. Boost to S' moving at $+\beta c$ along x — an observer running away from the source, in the same direction the light is going. Chapter 2.4 §2.1's boost matrix acts on any four-vector, so

$$\begin{aligned} k'^0 &= \gamma\left(k^0 - \beta k^1\right) = \gamma \frac{\omega}{c}\left(1-\beta\right), \\ k'^1 &= \gamma\left(k^1 - \beta k^0\right) = \gamma \frac{\omega}{c}\left(1-\beta\right). \end{aligned} \tag{2.5.54}$$

Both components pick up the same factor — as they must, since k'^μ has to be null too. Reading off $\omega' = ck'^0$ and simplifying with $\gamma = 1/\sqrt{(1-\beta)(1+\beta)}$:

$$\omega' \;=\; \frac{1-\beta}{\sqrt{(1-\beta)(1+\beta)}}\,\omega \;=\; \boxed{\sqrt{\frac{1-\beta}{1+\beta}}\,\omega} \tag{2.5.55}$$

Receding ($\beta > 0$): $\omega' < \omega$, redshift. Approaching: replace $\beta \to -\beta$ and get blueshift. At $\beta \ll 1$ this reduces to $\omega' \approx \omega(1-\beta)$, the classical result, so nothing familiar is lost. What is new is the exactness — and the second-order term, which is where the physics is.

7.3 · Transverse Doppler: time dilation, seen directly

Now a case with no classical counterpart at all. Let a source move at $+\beta c$ along x past a detector, and consider the light that reaches the detector travelling in the $+y$ direction *in the detector's frame* — the light emitted, as the lab sees it, at the moment of closest approach, so that the source's velocity is entirely perpendicular to the line of sight. Classically there is no Doppler shift whatsoever in that configuration: the distance is momentarily not changing.

In the lab frame S, the received wave has

$$k^\mu \;=\; \frac{\omega}{c}\left(1,\ 0,\ 1,\ 0\right). \tag{2.5.56}$$

Transform to the source's rest frame S', which moves at $+\beta c$ along x:

$$k'^0 \;=\; \gamma\left(k^0 - \beta k^1\right) \;=\; \gamma\frac{\omega}{c} - 0 \;=\; \gamma\,\frac{\omega}{c}. \tag{2.5.57}$$

But ck'^0 is the frequency in the source's own frame, which is the frequency the source actually emits — call it ω_0, a property of the atom. So $\omega_0 = \gamma\omega$, i.e.

$$\boxed{\omega \;=\; \frac{\omega_0}{\gamma}.} \tag{2.5.58}$$

A redshift, of exactly the factor γ, in a geometry where the classical prediction is precisely nothing. And its content is unmistakable: $1/\gamma$ is the time-dilation factor of Chapter 2.2. The source is a clock; it is moving; it runs slow; you see its emission slowed. **The transverse Doppler shift is time dilation with no admixture of anything else**, which is exactly why it was worth measuring.

▼ **Grind box — the general Doppler formula, aberration, and what Ives and Stilwell actually measured**

Arbitrary angle. Let the light travel at angle θ to the x-axis in S, so $k^\mu = (\omega/c)(1, \cos\theta, \sin\theta, 0)$. Boosting along x by β:

$$\begin{aligned} \omega' &= \gamma\omega\big(1 - \beta\cos\theta\big), \\ k'^1 &= \gamma\frac{\omega}{c}\big(\cos\theta - \beta\big), \qquad k'^2 = \frac{\omega}{c}\sin\theta. \end{aligned}$$

The first line is the **general Doppler formula**. Check the special cases: $\theta = 0$ gives $\omega' = \gamma\omega(1-\beta)$, which is (2.5.55) ✓; $\theta = \pi/2$ gives $\omega' = \gamma\omega$, which is (2.5.58) ✓ (with $\omega' = \omega_0$).

Aberration. Divide the second line by the first to get the direction in S':

$$\cos\theta' = \frac{k'^1}{k'^0} = \frac{\cos\theta - \beta}{1 - \beta\cos\theta},$$

which is the relativistic **aberration of light** — the same formula that turns the rain on your windscreen into a forward-slanting streak, but exact, and with the crucial difference that the aberration of light does not depend on any medium. This is Problem 3, where it becomes the headlight effect.

⚑ **Ives and Stilwell, 1938 — quoted experiment, derived analysis.** The transverse configuration is hard to arrange because "transverse" must be specified in a definite frame and a tiny angular error contaminates the result with a first-order longitudinal shift, which is $\gamma - 1 \approx \beta^2/2$ times *larger* in the relevant sense. Ives and Stilwell got around this by measuring longitudinally in both directions at once. Let a source recede with β and approach with β; in wavelengths, $\lambda_\pm = \lambda_0\sqrt{(1\pm\beta)/(1\mp\beta)}$. Average them:

$$\frac{\lambda_+ + \lambda_-}{2} = \frac{\lambda_0}{2}\left[\sqrt{\frac{1+\beta}{1-\beta}} + \sqrt{\frac{1-\beta}{1+\beta}}\right] = \frac{\lambda_0}{2}\cdot\frac{(1+\beta)+(1-\beta)}{\sqrt{(1-\beta)(1+\beta)}} = \gamma\,\lambda_0.$$

The first-order shifts cancel *exactly* and what survives is pure γ — the transverse effect, extracted from a longitudinal measurement. Ives and Stilwell ran hydrogen canal rays at $\beta \approx 5\times10^{-3}$, where $\gamma - 1 \approx 1.25\times10^{-5}$, and found the predicted second-order displacement of the mean. It remains one of the cleanest direct tests of time dilation, and the derivation above is the entire theory of the experiment. (We quote the experiment; the algebra is ours.)

A FOUR-VECTOR WAITING TO BE IDENTIFIED

Two null four-vectors have now appeared in this chapter for entirely different reasons: p^μ for a massless particle (§4.3), and k^μ for a light wave (§7.1). Nothing so far connects them. But notice that they have the same transformation law, the same null condition, and — if you compare (2.5.33) with $\omega = c\,|\mathbf{k}|$ — the same relation between time and space components.

Chapter 4.1 supplies the missing constant. The relation is

$$p^\mu = \hbar\, k^\mu, \qquad \text{i.e.} \qquad E = \hbar\omega, \quad \mathbf{p} = \hbar\mathbf{k},$$

and the reason it must be exactly this, with a single constant and no angle-dependent fudge, is that both sides are four-vectors: *one* scalar relates them, or none does. So the relativity of this chapter forces $E = \hbar\omega$ and $\mathbf{p} = \hbar\mathbf{k}$ to stand or fall together — you cannot have the Planck relation for energy without the de Broglie relation for momentum. That is a strong structural statement, and it was available years before anyone believed either half of it.

8 · Collisions: invariant mass, and why colliders exist

The whole practical value of §2 is that it turns collision problems into linear algebra. You write down the four-momenta, you add them, you square whatever combination kills the unknowns, and you are done. This section builds the three tools and then uses them to say something quantitative and slightly outrageous about accelerator design.

8.1 · The invariant mass of a system

Take any collection of particles, whatever they are doing, and add up their four-momenta:

$$P^\mu \;\equiv\; \sum_i p_i^\mu \;=\; \left(\frac{\sum_i E_i}{c},\; \sum_i \mathbf{p}_i \right). \tag{2.5.59}$$

A sum of four-vectors is a four-vector, so P^μ transforms properly and has an invariant square. Define the system's **invariant mass** M by

$$M^2c^2 \;\equiv\; P\cdot P \;=\; \frac{\left(\sum_i E_i\right)^2}{c^2} - \left|\sum_i \mathbf{p}_i\right|^2. \tag{2.5.60}$$

Everyone agrees on M. And if the system's total four-momentum is conserved — which by §2 it is — then M is conserved as well, which is a strong constraint on what a system may turn into.

8.2 · The centre-of-momentum frame

Suppose P^μ is timelike, $P \cdot P > 0$. Then by exactly the construction of Chapter 2.3 §6.1 there is a frame in which its spatial part vanishes: boost with $\beta_{\text{cm}} = \mathbf{P}c/P^0 = c\sum_i \mathbf{p}_i / \sum_i E_i$, which has magnitude less than 1 precisely because P^μ is timelike. In that frame — the **centre-of-momentum frame**, or CM frame —

$$P^\mu\Big|_{\text{CM}} = \left(\frac{E_{\text{CM}}}{c},\, \mathbf{0}\right), \qquad \text{so} \qquad P \cdot P = \frac{E_{\text{CM}}^2}{c^2} \quad \Longrightarrow \quad E_{\text{CM}} = Mc^2. \tag{2.5.61}$$

So the invariant mass of a system is its total energy in the frame where it is collectively at rest, divided by c^2. Notice that this is the direct generalisation of $E_0 = mc^2$ from one particle to many — and that β_{cm} is exactly (2.5.32) applied to the system's four-momentum, which is a small piece of evidence that we defined things sensibly.

⚠ **WHY THIS ISN'T OBVIOUS — MASS IS NOT A SUBSTANCE**

Take two photons of energy E flying in *opposite* directions along x. Each is massless. Add their four-momenta:

$$P^\mu = \frac{E}{c}(1, 1, 0, 0) + \frac{E}{c}(1, -1, 0, 0) = \left(\frac{2E}{c},\, 0,\, 0,\, 0\right),$$

so $P \cdot P = 4E^2/c^2 > 0$ and the pair has invariant mass $M = 2E/c^2$ — nonzero, built entirely out of massless constituents. There is even a CM frame: it is the one you are already in, since $\mathbf{P} = 0$.

Now aim the same two photons in the *same* direction. Then $P^\mu = (2E/c)(1, 1, 0, 0)$, $P \cdot P = 0$, and $M = 0$. Same two objects, same energies, different invariant mass.

So mass is not stuff that the parts have and the whole inherits. It is a property of the total four-momentum — specifically, of how much the constituent four-momenta point in *different* directions in spacetime. Randomly directed momentum inside a box shows up as mass of the box. This is not an analogy or a special case: it is where most of your own mass comes from, as §9 explains, and it is why "matter is made of massive things" is the wrong picture at the bottom.

8.3 · Mandelstam s

For a two-body collision the invariant mass of the initial state gets its own name. Define

$$s \;\equiv\; \big(p_1 + p_2\big) \cdot \big(p_1 + p_2\big)\; c^2, \qquad \text{so} \qquad \sqrt{s} \;=\; E_{\text{CM}}. \tag{2.5.62}$$

(The factor of c^2 is bookkeeping so that $\sqrt{s}$ is an energy in SI units. In the natural units of Part V, where $c = 1$, the definition is simply $s = (p_1 + p_2)^2$ and this parenthesis is unnecessary. Chapter 5.7 introduces the companions t and u; here we need only s.)

Expand it once, using $p_i \cdot p_i = m_i^2 c^2$:

$$\frac{s}{c^2} \;=\; p_1\cdot p_1 + 2\,p_1\cdot p_2 + p_2\cdot p_2 \;=\; \big(m_1^2+m_2^2\big)c^2 + 2\,p_1\cdot p_2. \tag{2.5.63}$$

Everything now depends on the single cross term, and $p_1 \cdot p_2$ is an invariant we can evaluate in whichever frame is convenient. Two configurations matter.

Fixed target. Particle 1 has energy E and hits particle 2 at rest, $p_2^\mu = (m_2 c, \mathbf{0})$. Then $p_1 \cdot p_2 = (E/c)(m_2 c) - \mathbf{p}_1 \cdot \mathbf{0} = m_2 E$, so

$$s_{\text{fixed}} \;=\; \big(m_1^2+m_2^2\big)c^4 \;+\; 2\,m_2c^2\,E \qquad \xrightarrow[E\gg mc^2]{} \qquad \sqrt{s} \;\approx\; \sqrt{2m_2c^2E}. \tag{2.5.64}$$

Collider. Two beams of energy E meet head-on with equal and opposite momenta. Then $\sum \mathbf{p}_i = 0$, so you are already in the CM frame and

$$\sqrt{s_{\text{collider}}} \;=\; E_{\text{CM}} \;=\; 2E. \tag{2.5.65}$$

THE WHOLE ARGUMENT FOR BUILDING COLLIDERS, IN ONE LINE

Compare the two scalings. Doubling the beam energy of a collider doubles $\sqrt{s}$. Doubling the beam energy of a fixed-target machine multiplies $\sqrt{s}$ by $\sqrt{2}$.

$$\sqrt{s_{\text{collider}}} \propto E, \qquad\qquad \sqrt{s_{\text{fixed}}} \propto \sqrt{E}.$$

Useful energy is $\sqrt{s}$, not E — the rest, in the fixed-target case, is spent dragging the centre of mass forward and is unavailable for making anything. The consequence compounds viciously with energy, and Worked Example 2 puts a number on it that is hard to believe until you check it.

8.4 · Threshold energies, and the antiproton

Here is the technique that makes s worth defining. A reaction can proceed only if the initial state carries enough invariant mass to build the final state. The *minimum* case is unambiguous: at threshold, the products have no kinetic energy left over in the CM frame — they are all at rest there, moving together — so the final invariant mass is just the sum of the final rest masses. Since s is conserved and invariant,

$$\sqrt{s}\,\Big|_{\text{threshold}} \;=\; \Big(\sum_{\text{final}} m_f\Big)\,c^2. \tag{2.5.66}$$

Take the reaction that mattered historically: producing an antiproton by smashing a proton beam into a hydrogen target,

$$p + p \;\longrightarrow\; p + p + p + \bar{p}. \tag{2.5.67}$$

Charge and baryon number both force the extra proton to accompany the antiproton — you cannot make $\bar{p}$ alone — so four particles of mass m_p must emerge. (Antiparticles have the same rest mass as their partners; ⚑ we quote that here, and Chapter 5.4 derives it, where it falls out of the Dirac equation.) Setting $\sqrt{s} = 4m_pc^2$ in (2.5.64) with $m_1 = m_2 = m_p$:

$$\begin{aligned} 16\, m_p^2 c^4 &= 2m_p^2 c^4 + 2m_p c^2 E \\ \Longrightarrow \qquad E &= \frac{14\, m_p^2 c^4}{2m_p c^2} = 7\, m_p c^2, \end{aligned} \tag{2.5.68}$$

so the projectile's *kinetic* energy at threshold is

$$T = E - m_p c^2 = 6\, m_p c^2 = 6 \times 938.27 \text{ MeV} = 5.63 \text{ GeV}. \tag{2.5.69}$$

Six proton rest energies to make two particles' worth of new rest mass — the factor of three overhead is exactly the energy locked up in the forward motion of the centre of mass, which (2.5.64) says you cannot spend. ⚑ Historically: the Bevatron at Berkeley was designed to reach $6.2\ \text{GeV}$, comfortably above this threshold and for this reason, and the antiproton was found there in 1955. The number in (2.5.69) is a piece of accelerator engineering that came out of four lines of four-vector algebra.

▼ **Grind box — the general threshold formula, and a second worked case**

Do the algebra once in general so you never have to repeat it. A projectile of mass m_1 and total energy E strikes a stationary target of mass m_2 and produces final particles of total rest mass $\mathcal{M} \equiv \sum_f m_f$. Equate (2.5.64) with (2.5.66):

$$(m_1^2 + m_2^2)c^4 + 2m_2 c^2 E \;=\; \mathcal{M}^2 c^4,$$

$$\boxed{E_{\text{threshold}} \;=\; \frac{\left[\mathcal{M}^2 - m_1^2 - m_2^2\right] c^2}{2m_2}.}$$

Check against (2.5.68): $\mathcal{M} = 4m_p, m_1 = m_2 = m_p$ gives $E = (16 - 1 - 1)m_pc^2/2 = 7m_pc^2$ ✓.

A second case — pion photoproduction, $\gamma + p \to p + \pi^0$. Now $m_1 = 0$ (a photon), $m_2 = m_p = 938.27\ \text{MeV}/c^2$, and $\mathcal{M} = m_p + m_{\pi^0}$ with $m_{\pi^0}c^2 = 134.98\ \text{MeV}$ (⚑ quoted mass). Then

$$E_\gamma = \frac{(m_p + m_\pi)^2 - m_p^2}{2m_p} c^2 = \frac{2m_p m_\pi + m_\pi^2}{2m_p} c^2 = m_\pi c^2 \left(1 + \frac{m_\pi}{2m_p}\right).$$

Numerically $134.98 \times (1 + 0.0719) = 144.7\ \text{MeV}$. The photon must supply about 7% more than the pion's rest energy, the surplus being the recoil the proton is obliged to take. Note how the formula degrades gracefully: for a very heavy target, $m_2 \to \infty$ and the overhead vanishes, which is why a nucleus makes a better anvil than a single proton.

The same physics, from the CM frame. There is a shortcut worth knowing. The overhead in fixed-target work is the kinetic energy of the centre of mass, which by (2.5.32) moves at $\beta_{\text{cm}} = p_1 c/(E + m_2 c^2)$. Everything above can be re-derived by boosting to that frame and demanding the products be at rest, but the invariant route is shorter precisely because it never needs β_{cm} at all. That is the habit to acquire: *find the invariant, evaluate it in the easy frame, use it in the hard one.*

9 · Mass is not additive

Section 2.3's grind box already produced the result that makes this section necessary: two lumps of putty of rest mass m each, colliding and sticking, form a body of rest mass $2\gamma m$, not $2m$. The kinetic energy did not disappear — it became rest mass. Run that backwards and you have the most consequential fact in applied physics.

9.1 · Binding energy and the mass defect

Consider a bound system: a nucleus, an atom, a planet. Assemble it from constituents that start far apart and at rest, and let it settle by radiating away the excess. The total four-momentum is conserved throughout, so in the frame where the final object sits at rest,

$$\Big(\sum_i m_i\Big)c^2 \;=\; M_{\text{bound}}\,c^2 \;+\; B, \qquad \text{i.e.} \qquad \boxed{M_{\text{bound}} = \sum_i m_i - \frac{B}{c^2}} \tag{2.5.70}$$

where $B > 0$ is the energy carried away — the **binding energy**. The bound system is *lighter than its parts*, by exactly the energy you would have to put back to take it apart, over c^2. That difference is the **mass defect**, and it is not a small correction to a picture in which mass is additive; it is a demonstration that mass was never additive.

Real numbers, since the whole point is that this is measurable. ⚑ The masses below are quoted from standard tables; everything done with them is ours.

SYSTEM	CONSTITUENTS (MeV/c^2)	BOUND MASS	B	$B/\sum mc^2$
Deuteron $^2\mathrm{H}$	$938.272 + 939.565 = 1877.837$	1875.613	2.224 MeV	0.118%
Helium-4	$2(938.272) + 2(939.565) = 3755.675$	3727.379	28.30 MeV	0.753%
Hydrogen atom	$938.272 + 0.511 = 938.783$	938.783	13.6 eV	1.4×10^{-8}

Read the last column. Chemistry runs at 10^{-8} or 10^{-10} of the rest energy; that is why Lavoisier could weigh reactants and products and conclude that mass is conserved — his balance was many orders of magnitude too coarse to see the defect, and he was right to the precision available. Nuclear binding runs at 10^{-3}, five to seven orders of magnitude larger, which is the entire reason nuclear energy is a different kind of thing rather than a better kind of chemistry.

▼ **Grind box — the arithmetic, and a fusion reaction end to end**

Deuteron. A proton and a neutron bound together. Constituent rest energies $938.272 + 939.565 = 1877.837$ MeV; measured deuteron rest energy 1875.613 MeV. Difference:

$$B = 1877.837 - 1875.613 = 2.224 \text{ MeV}, \qquad \frac{B}{\sum mc^2} = \frac{2.224}{1877.837} = 1.184 \times 10^{-3}.$$

So a deuteron weighs about 0.12% less than the parts you made it from — a fractional mass difference roughly 10^5 times larger than any chemical bond and comfortably measurable with a mass spectrometer.

Helium-4. $2 \times 938.272 + 2 \times 939.565 = 3755.675$; measured 3727.379; so $B = 28.296$ MeV, i.e. 7.07 MeV per nucleon and 0.753% of the total. Helium-4 is unusually tightly bound for its size, which is why α particles exist as a distinct thing and why stellar nucleosynthesis piles up at helium.

A complete reaction: D–T fusion. ${}^2\mathrm{H} + {}^3\mathrm{H} \to {}^4\mathrm{He} + n$. Using nuclear rest energies in MeV:

$$\begin{aligned} \text{in:} \quad & 1875.613 + 2808.921 = 4684.534 \\ \text{out:} \quad & 3727.379 + 939.565 = 4666.944 \\ \text{released:} \quad & Q = 4684.534 - 4666.944 = 17.59 \text{ MeV}. \end{aligned}$$

As a fraction of the input rest energy, $17.59/4684.534 = 3.755 \times 10^{-3}$, i.e. 0.375%. Per kilogram of fuel that is $0.00375\, c^2 = 3.4 \times 10^{14}$ J kg^{-1}, against roughly 5×10^7 J kg^{-1} for burning hydrocarbons — a factor of about seven million.

The lesson to extract. Every one of these is the same calculation: add the rest energies before, add them after, and the difference is kinetic energy. Nothing was "converted into energy" in a way that requires new physics. The bookkeeping quantity $\sum_i E_i$ was conserved throughout, and all that changed was how much of it was sitting in the rest-mass column.

⚠ WHY THIS ISN'T OBVIOUS — "MASS CONVERTS TO ENERGY" IS THE WRONG SENTENCE

The standard telling is that in a nuclear reaction "some mass is converted into energy, according to $E = mc^2$". Nothing in this chapter supports that reading, and it causes real confusion.

What is actually true. The conserved quantity is $E = \sum_i \gamma_i m_i c^2$, and it never changes. In a fusion reaction the rest masses of the products are smaller than those of the reactants, so the *rest-energy* share of the total falls; the difference reappears in the kinetic-energy share. Nothing was created or destroyed and nothing turned into anything. Energy was reallocated between two columns of the same ledger.

And the system's mass? If you do the reaction inside a sealed, perfectly reflecting box and weigh the box, its invariant mass is *unchanged* — because the kinetic energy and radiation are still inside, and by §8.2 the box's invariant mass is its total energy in its rest frame over c^2. The mass only drops once you let the heat out. So the honest sentence is: *rest energy is released as kinetic energy, and if the kinetic energy escapes, the system's rest mass drops by exactly the energy that left, over c^2.*

A related trap. "$E = mc^2$" is about rest energy. The energy of a *moving* body is $E = \gamma mc^2$, and the two differ by every factor that matters at an accelerator. When a physicist writes $E = mc^2$ they mean the $v = 0$ case of (2.5.26), and the famous equation is famous partly because that qualification is usually dropped.

9.2 · Where your mass actually comes from

Push the argument one step further than nuclear physics does and it stops being a correction and becomes the main effect.

A proton has rest energy $938\ \mathrm{MeV}$. It is made of three light quarks whose rest energies sum to roughly $9\ \mathrm{MeV}$ — ⚑ quoted from Chapter 6.5's tables, and note that even *defining* a quark mass takes care, since they are never found alone. That accounts for about 1% of the proton. The other 99% is exactly what §8.2's warning described: the energy of quarks moving relativistically inside a small volume, plus the energy of the gluon field binding them, all of it appearing as invariant mass because the constituent four-momenta point in many different directions and their spatial parts cancel while their energies add.

So the mass of ordinary matter is, to better than 99%, not a property of its ingredients. It is confined kinetic and field energy, measured in the frame where the total momentum vanishes — which is (2.5.60), applied to a bound state of a strongly coupled field theory. Chapter 6.5 does that calculation, or rather explains why it can only be done numerically; the structural statement is available now, and it is this chapter's, not quantum chromodynamics'.

10 · Worked examples

WORKED EXAMPLE 1 — COMPTON SCATTERING, IN FULL

A photon of wavelength λ strikes a free electron at rest. The photon scatters through an angle θ and emerges with wavelength λ'; the electron recoils in some unknown direction with some unknown speed. Find $\lambda' - \lambda$.

Why this is the model calculation. There are four unknowns after the collision — the photon's new wavelength, and the electron's three momentum components — and only four conservation equations. But we are not asked about the electron at all. The technique is therefore: *isolate the four-momentum you do not care about on one side, and square it*, because squaring converts an unknown four-vector into the one number you do know about it, namely m^2c^2. That single move removes the recoil direction from the problem without ever computing it.

Setup. ⚑ One quoted input, and only one: a photon of wavelength λ has energy $E = hc/\lambda$. That is Planck and Einstein, and Chapter 4.1 is where it is argued for; everything else below is this chapter's. Given it, §4.3's $E = pc$ gives the photon's momentum $|\mathbf{p}| = h/\lambda$, so with $\hat{n}$ a unit vector along its travel,

$$p_\gamma^\mu = \frac{h}{\lambda}\big(1,\ \hat{n}\big), \qquad p_\gamma'^\mu = \frac{h}{\lambda'}\big(1,\ \hat{n}'\big), \qquad \hat{n}\cdot\hat{n}' = \cos\theta,$$

and the electron starts at rest, $p_e^\mu = (m_e c,\ \mathbf{0})$, with unknown final $p_e'^\mu$.

Step 1 — conservation, rearranged to isolate the unknown.

$$p_\gamma^\mu + p_e^\mu = p_\gamma'^\mu + p_e'^\mu \qquad \Longrightarrow \qquad p_e'^\mu = p_\gamma^\mu - p_\gamma'^\mu + p_e^\mu.$$

Step 2 — square both sides. The left side is known: $p_e'\cdot p_e' = m_e^2c^2$, by (2.5.29), whatever the electron ended up doing. Expanding the right side needs six terms, so take them one at a time:

$$\begin{aligned} p_e'\cdot p_e' &= \underbrace{p_\gamma\cdot p_\gamma}_{=0} + \underbrace{p_\gamma'\cdot p_\gamma'}_{=0} + \underbrace{p_e\cdot p_e}_{=m_e^2c^2} \\ &\quad - 2\,p_\gamma\cdot p_\gamma' + 2\,p_\gamma\cdot p_e - 2\,p_\gamma'\cdot p_e. \end{aligned}$$

The first two vanish because photons are massless. Setting the whole thing equal to $m_e^2c^2$, the $m_e^2c^2$ terms cancel and we are left with a relation among three dot products:

$$p_\gamma\cdot p_\gamma' \;=\; p_\gamma\cdot p_e \;-\; p_\gamma'\cdot p_e.$$

The electron's final state has vanished entirely. That is the whole trick.

Step 3 — evaluate the three dot products. Each is elementary; keep the metric signs straight.

$$p_\gamma\cdot p_\gamma' = \frac{h}{\lambda}\frac{h}{\lambda'}\big(1\cdot 1 - \hat{n}\cdot\hat{n}'\big) = \frac{h^2}{\lambda\lambda'}\big(1-\cos\theta\big),$$

$$p_\gamma\cdot p_e = \frac{h}{\lambda}\,m_e c\,(1) - \frac{h}{\lambda}\hat{n}\cdot\mathbf{0} = \frac{h\,m_e c}{\lambda}, \qquad p_\gamma'\cdot p_e = \frac{h\,m_e c}{\lambda'}.$$

Step 4 — assemble.

$$\frac{h^2}{\lambda\lambda'}\big(1-\cos\theta\big) \;=\; h\,m_e c\left(\frac{1}{\lambda}-\frac{1}{\lambda'}\right) \;=\; h\,m_e c\,\frac{\lambda'-\lambda}{\lambda\lambda'}.$$

Multiply through by $\lambda\lambda'/(h\,m_e c)$ — the product $\lambda\lambda'$, which was the only place the two wavelengths appeared together, cancels completely:

$$\boxed{\lambda'-\lambda \;=\; \frac{h}{m_e c}\big(1-\cos\theta\big).}$$

What the answer says. The shift does not depend on λ. It is bounded, between 0 at $\theta=0$ and $2h/m_e c$ at $\theta=180^\circ$. And it is a pure length times a pure number.

The Compton wavelength, and a payoff from Chapter 0.3. The combination

$$\lambda_C \equiv \frac{h}{m_e c} = \frac{6.626\times10^{-34}}{(9.109\times10^{-31})(2.998\times10^{8})} = 2.426\times10^{-12}\ \mathrm{m} = 2.426\ \mathrm{pm}$$

is the *only* length you can build from h, m_e and c — check it against Chapter 0.3 §5's method: with $[h]=\mathsf{ML}^2\mathsf{T}^{-1}$, $[m_e]=\mathsf{M}$, $[c]=\mathsf{LT}^{-1}$, seeking $h^A m_e^B c^C$ of dimension L gives $A+B=0, 2A+C=1, -A-C=0$, whence $A=1, C=-1, B=-1$, uniquely. So dimensional analysis alone guaranteed that *if* the answer is a wavelength shift built from these three constants, it must be λ_C times a dimensionless function of θ. The four-vector calculation supplied the function: $1-\cos\theta$.

Numbers, and why anyone believed it. Compton used molybdenum K_α X-rays, $\lambda=71\ \mathrm{pm}$. At $\theta=90^\circ$ the shift is $2.426\ \mathrm{pm}$ — a 3.4% change, easily resolved. At $\theta=180^\circ$ it is $4.853\ \mathrm{pm}$, or 6.8%. Crucially, the shift is *independent of the incident wavelength and of the target material*, which is impossible for any classical scattering mechanism: a classical wave shakes an electron at the driving frequency and the electron re-radiates at that same frequency, giving no shift at all. Chapter 4.1 uses exactly this calculation as evidence that light carries momentum in discrete parcels obeying (2.5.33) — the collision is between two *particles*, and the bookkeeping that works is this chapter's.

WORKED EXAMPLE 2 — WHAT A FIXED-TARGET LHC WOULD COST

The LHC collides protons head-on at $6.8\ \mathrm{TeV}$ *per beam, giving* $\sqrt{s} = 13.6\ \mathrm{TeV}$*. What beam energy would a fixed-target machine need to reach the same* $\sqrt{s}$ *against a stationary proton? And conversely, what does the LHC's own beam achieve if you point it at a stationary target instead?* (⚑ The machine parameters are quoted; the physics is ours.)

Part (a) — the fixed-target equivalent. Rearrange (2.5.64) with $m_1 = m_2 = m_p$:

$$E = \frac{s - 2m_p^2c^4}{2m_pc^2}.$$

With $\sqrt{s} = 13.6\ \mathrm{TeV} = 1.36 \times 10^4\ \mathrm{GeV}$ and $m_pc^2 = 0.9383\ \mathrm{GeV}$, the $2m_p^2c^4 = 1.76\ \mathrm{GeV}^2$ is utterly negligible against $s = 1.85 \times 10^8\ \mathrm{GeV}^2$, so

$$E \approx \frac{1.850 \times 10^8}{2 \times 0.9383}\ \mathrm{GeV} = 9.86 \times 10^7\ \mathrm{GeV} = 9.86 \times 10^4\ \mathrm{TeV} \approx 99\ \mathrm{PeV}.$$

Compare with the $6.8\ \mathrm{TeV}$ each LHC beam actually carries: the fixed-target machine would need

$$\frac{9.86 \times 10^7}{6.8 \times 10^3} \approx 1.45 \times 10^4$$

times more energy per particle — about fourteen and a half thousand. The LHC's ring is $27\ \mathrm{km}$ around and its bending magnets are near the limit of what superconductors will do; scaling the same technology by 1.45×10^4 means a ring on the order of $4 \times 10^5\ \mathrm{km}$, which is about the distance to the Moon. That is the $\sqrt{E}$ scaling, made visceral.

Part (b) — the LHC beam on a fixed target. Same formula, other direction, with $E = 6.8 \times 10^3\ \mathrm{GeV}$:

$$\sqrt{s} = \sqrt{2m_pc^2E + 2m_p^2c^4} = \sqrt{2(0.9383)(6800) + 1.76}\ \ \mathrm{GeV} = 113\ \mathrm{GeV}.$$

So a $6.8\ \mathrm{TeV}$ proton hitting a stationary proton makes only $113\ \mathrm{GeV}$ of invariant mass available — **1.7% of the beam energy**. The other 98.3% is spent dragging the centre of mass down the beam pipe and is gone.

The overhead is not a constant. Measure the waste by the ratio of beam energy to available energy, $E/\sqrt{s}$. Rearranging (2.5.64) for equal masses gives

$$\frac{E}{\sqrt{s}} = \frac{\sqrt{s}}{2m_pc^2} - \frac{m_pc^2}{\sqrt{s}} \xrightarrow[\sqrt{s} \gg m_pc^2]{} \frac{\sqrt{s}}{2m_pc^2},$$

which *grows linearly with the energy you are trying to reach*. At the antiproton threshold of §8.4, $\sqrt{s} = 4m_pc^2 = 3.75\ \mathrm{GeV}$ and the ratio is $2 - \frac{1}{4} = 1.75$: you buy $3.75\ \mathrm{GeV}$ of invariant mass with a $6.57\ \mathrm{GeV}$ beam, which is tolerable. At the LHC's beam energy the same ratio is $6800/113 = 60$. There is no fixed inefficiency to design around — the waste compounds — which is why every energy-frontier machine built since the 1970s has been a collider, and why fixed-target experiments now go looking for rare processes at modest $\sqrt{s}$ rather than for new mass.

11 · Your turn

PROBLEM 1 — A PHOTON CANNOT DECAY INTO AN ELECTRON–POSITRON PAIR

In empty space, consider the proposed process $\gamma \to e^- + e^+$. Show, using invariant mass alone, that it is impossible — no matter how energetic the photon is. Then explain in one sentence why the same process *does* occur routinely near a heavy nucleus.

– Solution

Four-momentum conservation would require $p_\gamma^\mu = p_-^\mu + p_+^\mu$. Squaring both sides is legitimate because both are four-vectors, and the square is an invariant, so if the two sides are equal their squares are equal in every frame.

Left side. The photon is massless: $p_\gamma \cdot p_\gamma = 0$.

Right side. Let $P^\mu = p_-^\mu + p_+^\mu$. Each electron four-momentum is timelike and future-pointing (positive energy), so their sum is too, and by §8.2 there is a frame in which $\mathbf{P} = 0$. Evaluate the invariant *there*, which is the whole art:

$$P \cdot P\Big|_{\text{CM}} = \frac{(E_- + E_+)^2}{c^2} - 0.$$

In that frame the two particles have equal and opposite momenta, so equal energies, and each satisfies $E_\pm \geq m_e c^2$ by (2.5.31). Hence

$$P \cdot P \;\geq\; \frac{(2m_e c^2)^2}{c^2} \;=\; 4m_e^2 c^2 \;>\; 0.$$

So the process demands $0 = P \cdot P \geq 4m_e^2 c^2$, a contradiction. The photon's energy never entered the argument, because $P \cdot P$ is an invariant and we were free to compute it in the pair's own rest frame. ■

Restated geometrically, which is the version to remember: a null vector cannot be the sum of two timelike future-pointing vectors. The sum of future-timelike vectors is future-timelike, strictly — the light cone is the boundary and you cannot get back onto it by adding things from the inside.

Near a nucleus. The nucleus absorbs four-momentum, so the reaction is really $\gamma + N \to e^- + e^+ + N$, and the initial invariant mass is no longer zero: by (2.5.63) it is $s = m_N^2 c^4 + 2m_N c^2 E_\gamma$, which exceeds $(m_N + 2m_e)^2 c^4$ once E_γ is large enough. Applying the grind box's threshold formula with $m_1 = 0$, $m_2 = m_N$, $\mathcal{M} = m_N + 2m_e$:

$$E_\gamma^{\text{thr}} = \frac{(m_N + 2m_e)^2 - m_N^2}{2m_N} c^2 = 2m_e c^2 \left(1 + \frac{m_e}{m_N}\right),$$

> which for a heavy nucleus is barely above $2m_ec^2 = 1.022\ \mathrm{MeV}$. The nucleus takes almost no energy but supplies the momentum balance — it is there to break the kinematics, not to pay for it. This is why $1.022\ \mathrm{MeV}$ is the number quoted for pair production, and why positron emission tomography works at $511\ \mathrm{keV}$ per photon in the reverse process.

PROBLEM 2 — THE RELATIVISTIC ROCKET, AND ITS ENERGY BUDGET

A ship accelerates with constant *proper* acceleration α — meaning the accelerometer bolted to its floor always reads α — starting from rest at $\tau = 0$.

(a) Using $u \cdot u = c^2, u \cdot a = 0$ and $a \cdot a = -\alpha^2$, show that $u^\mu = c(\cosh\phi, \sinh\phi, 0, 0)$ with $\phi = \alpha\tau/c$, and hence that $v/c = \tanh(\alpha\tau/c)$ and $\gamma = \cosh(\alpha\tau/c)$.

(b) With $\alpha = g = 9.81\ \mathrm{m\,s^{-2}}$, find v/c and γ after one year and after ten years of ship-time, and the energy per kilogram of payload required in each case.

− Solution

(a) The constraint $u \cdot u = c^2$ says the four-velocity is confined to a hyperbola in the (u^0, u^1) plane, and the general parametrisation of $X^2 - Y^2 = c^2$ is $X = c\cosh\phi, Y = c\sinh\phi$ for some function $\phi(\tau)$ — precisely Chapter 2.3 §3.2's hyperbolic parametrisation, which is why rapidity was worth defining. So $u^\mu = c(\cosh\phi, \sinh\phi, 0, 0)$ with no loss of generality. Differentiate:

$$a^\mu = \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = c\,\frac{\mathrm{d}\phi}{\mathrm{d}\tau}\big(\sinh\phi,\ \cosh\phi,\ 0,\ 0\big).$$

Check the orthogonality first, as a sanity test: $u \cdot a = c^2\dot\phi(\cosh\phi\sinh\phi - \sinh\phi\cosh\phi) = 0$ ✓, automatically — the constraint (2.5.13) is built into the parametrisation. Now the magnitude:

$$a \cdot a = c^2\dot\phi^2\big(\sinh^2\phi - \cosh^2\phi\big) = -c^2\dot\phi^2.$$

Setting this equal to $-\alpha^2$ gives $\dot\phi = \alpha/c$, so $\phi = \alpha\tau/c$ with $\phi(0) = 0$ for a start from rest. Finally, from $u^\mu = \gamma(c, \mathbf{v})$ we read off $\gamma = \cosh\phi$ and $\gamma v = c\sinh\phi$, hence

$$\frac{v}{c} = \tanh\left(\frac{\alpha\tau}{c}\right), \qquad \gamma = \cosh\left(\frac{\alpha\tau}{c}\right).$$

Constant proper acceleration is uniform growth of rapidity. That is the clean statement, and it is why tanh appears: velocities do not add, rapidities do, and a constant push adds rapidity at a constant rate. It also settles the speed limit question for good — tanh approaches 1 and never reaches it, however long you burn.

(b) The natural timescale is $c/g = (2.998 \times 10^8)/9.81 = 3.056 \times 10^7\ \mathrm{s} = 0.968$ yr. So one year of ship-time is $\phi = 1/0.968 = 1.033$.

SHIP-TIME τ	$\phi = g\tau/c$	$v/c = \tanh\phi$	$\gamma = \cosh\phi$	KINETIC ENERGY PER KG
1 yr	1.033	0.7750	1.582	5.23×10^{16} J
10 yr	10.33	$1 - 2 \times 10^{-9}$	1.53×10^4	1.37×10^{21} J

The kinetic energy is $E - mc^2 = (\gamma - 1)mc^2$, with $c^2 = 8.988 \times 10^{16}\ \mathrm{J\,kg^{-1}}$. After one year it is $0.582\,mc^2$ per kilogram — more than half the payload's rest energy, and that is *only* the payload, ignoring the fuel needed to carry the fuel. After ten years it is 1.5×10^4 times the rest energy: to deliver one kilogram you must supply the total annihilation energy of about fifteen tonnes, and no engine is perfect.

The point. Relativity does not forbid interstellar travel; it prices it. And it prices it exponentially, because $\gamma = \cosh\phi \sim \frac{1}{2}\mathrm{e}^{\phi}$ grows exponentially in ship-time, and so does the distance covered, $x = (c^2/g)(\cosh\phi - 1)$. Setting $x = 26\,000$ light-years — the galactic centre — gives $\cosh\phi = 2.68 \times 10^4$, hence $\phi = 10.89$ and $\tau = 10.5$ years of ship-time. That is the romance, and it is real. The bill is $(\gamma - 1)mc^2 \approx 2.7 \times 10^4\ mc^2$ per kilogram delivered — and since the ship never quite reaches c, some 2.6×10^4 years will have passed on Earth by the time it arrives.

PROBLEM 3 — THE HEADLIGHT EFFECT

A source at rest in frame S' emits photons uniformly in all directions. S' moves at βc along x relative to the lab. Using the aberration formula from §7's grind box, show that the photons emitted into the forward hemisphere in S' (those with $\theta' < 90^\circ$) are all compressed, in the lab, into a cone about the x-axis of half-angle $\theta_{1/2} = \arccos\beta$, and that for $\gamma \gg 1$ this is approximately $1/\gamma$. Evaluate for $\gamma = 10$ and $\gamma = 1000$, and comment on what this does to the apparent brightness of a relativistic jet pointed at you.

− Solution

The grind box in §7 derived $\cos\theta' = (\cos\theta - \beta)/(1 - \beta\cos\theta)$, mapping lab angle to source angle. We want the inverse, which is the same formula with $\beta \to -\beta$ (swapping the roles of the frames):

$$\cos\theta = \frac{\cos\theta' + \beta}{1 + \beta\cos\theta'}.$$

The boundary ray. Put $\theta' = 90^\circ$, i.e. $\cos\theta' = 0$: then $\cos\theta = \beta$. So the photon emitted exactly sideways in the source frame arrives in the lab at angle $\theta_{1/2} = \arccos\beta$ to the direction of motion. Since the map is monotonic in $\cos\theta'$, every ray with $\theta' < 90^\circ$ lands inside that cone.

The small-angle form. For $\theta_{1/2}$ small, $\sin\theta_{1/2} = \sqrt{1 - \cos^2\theta_{1/2}} = \sqrt{1-\beta^2} = 1/\gamma$, so $\theta_{1/2} \approx 1/\gamma$ radians. Neat, and worth remembering as a rule of thumb.

γ	β	$\theta_{1/2} = \arccos\beta$	SOLID ANGLE FRACTION
10	0.99499	$5.73^\circ = 0.100$ rad	$\approx 2.5 \times 10^{-3}$
1000	0.9999995	$0.0573^\circ = 1.0 \times 10^{-3}$ rad	$\approx 2.5 \times 10^{-7}$

(The solid-angle fraction of a cone of half-angle θ is $(1-\cos\theta)/2 \approx \theta^2/4 = 1/4\gamma^2$.)

Brightness. Half of all the emitted photons — the forward hemisphere, which is 2π steradians in the source frame — end up inside a lab cone of solid angle $\approx \pi/\gamma^2$. The photon flux per unit solid angle is therefore boosted by roughly γ^2, and each of those photons is *also* blueshifted by the longitudinal Doppler factor $\sqrt{(1+\beta)/(1-\beta)} \approx 2\gamma$ from (2.5.55), and they arrive at a compressed rate for the same reason. Multiplying the effects gives an apparent brightness enhanced by a large power of γ — the standard estimate for a continuously emitting jet is γ^4 in the observed flux, though the exact exponent depends on the source's spectral shape.

Consequence. A jet pointed within $1/\gamma$ of your line of sight looks enormously brighter than the identical jet pointed away, which is why blazars — active galaxies whose jets happen to aim at Earth — dominate the catalogues of the brightest gamma-ray sources despite being a tiny fraction of active galaxies. It is also why radio astronomers see one-sided jets from manifestly two-sided sources: the

> receding jet is de-beamed by the same large factor. Neither observation requires any asymmetry in the source. Both are (2.5.52), boosted.

PROBLEM 4 — MASS DEFECT AND ENERGY RELEASE IN D–D FUSION

Consider $^2\text{H} + {}^2\text{H} \to {}^3\text{He} + n$. Using nuclear rest energies $m_d c^2 = 1875.613$, $m_{^3\text{He}} c^2 = 2808.391$, $m_n c^2 = 939.565$ (all in MeV), compute (a) the energy released Q; (b) Q as a fraction of the input rest energy; (c) the energy released per kilogram of deuterium fuel, and compare with burning an equal mass of methane ($\approx 5.6 \times 10^7$ J kg^{-1}). (d) Finally, the neutron and the ^{3}He share Q as kinetic energy; use momentum conservation in the CM frame to find how it splits, and check that the non-relativistic treatment is justified.

− Solution

(a) Total four-momentum conservation, evaluated in the CM frame where the initial nuclei are brought together with negligible kinetic energy, gives $\sum_{\text{in}} mc^2 = \sum_{\text{out}} mc^2 + Q$:

$$\begin{aligned} \text{in:} \quad & 2 \times 1875.613 = 3751.226 \text{ MeV} \\ \text{out:} \quad & 2808.391 + 939.565 = 3747.956 \text{ MeV} \end{aligned}$$

$$Q = 3751.226 - 3747.956 = 3.270 \text{ MeV}.$$

(b) $Q / \sum_{\text{in}} mc^2 = 3.270/3751.226 = 8.72 \times 10^{-4}$, i.e. 0.0872% — under a tenth of a percent of the rest energy, and yet:

(c) Energy per kilogram is that fraction times c^2:

$$8.72 \times 10^{-4} \times 8.988 \times 10^{16} \text{ J kg}^{-1} = 7.8 \times 10^{13} \text{ J kg}^{-1}.$$

Against methane's 5.6×10^7, that is a factor of 1.4×10^6 — a million and a half. One kilogram of deuterium releases what about 1400 tonnes of methane would. The *entire* difference between chemistry and nuclear physics is the difference between the last column of §9.1's table for a hydrogen atom (10^{-8}) and for a nucleus (10^{-3}); the mechanism is identical and only the scale of the binding differs.

(d) In the CM frame the products have equal and opposite momenta, p. Non-relativistically the kinetic energies are $T_i = p^2/2m_i$, so they are inversely proportional to the masses:

$$\frac{T_n}{T_{^3\text{He}}} = \frac{m_{^3\text{He}}}{m_n} = \frac{2808.391}{939.565} = 2.989.$$

With $T_n + T_{^3\text{He}} = Q = 3.270$ MeV, this gives

$$T_n = Q\,\frac{2.989}{3.989} = 2.451 \text{ MeV}, \qquad T_{^3\text{He}} = 0.819 \text{ MeV}.$$

The light one takes most of the energy — a general feature of two-body decay, and the reason fusion reactors must cope with fast neutrons.

Justifying the approximation. The neutron's kinetic energy is $2.451/939.565 = 2.6 \times 10^{-3}$ of its rest energy, so $\gamma = 1.0026$ and $\beta = 0.072$. By the grind box in §4, the fractional error in using $T = p^2/2m$ at this speed is about $x^4/8$ with $x = \gamma\beta = 0.0723$, i.e. 3×10^{-6} — three parts per million, far below the precision of the input masses. The non-relativistic split is fine. Note that we needed relativity to compute Q at all, and then did not need it to divide Q up; that combination is typical of nuclear physics.

THE BRICK YOU JUST LAID

You have mechanics that survives a boost. The route was one demand — **a conservation law must be an equation between tensors** — and everything else was consequence: differentiate by the invariant τ to get u^μ with $u \cdot u = c^2$; multiply by m to get p^μ; observe that conserving three components of a four-vector in every frame forces the fourth, and that the fourth expands to $mc^2 + \frac{1}{2}mv^2 + \frac{3}{8}mv^4/c^2 + \cdots$. From the invariant square came $E^2 = p^2c^2 + m^2c^4$, from that the massless case as the null case, and from $\mathbf{v} = \mathbf{p}c^2/E$ the fact that null four-momentum means exactly c.

You also have the action. $S = -mc^2 \int \mathrm{d}\tau$ was *solved for*, not guessed, and it closed Chapter 2.3's maximisation theorem: extremising the action and maximising proper time are the same instruction, differing by the minus sign that converts one into the other. That is entry two in Chapter 1.2 §8.1's table, now derived.

And you have the tools that make relativistic problems tractable: contract to kill unknowns, boost to the frame where the answer is easy, and remember that invariant mass is a property of a four-momentum rather than a substance carried by parts.

Where this gets spent. Chapter 2.6 takes $\mathrm{d}p^\mu/\mathrm{d}\tau = f^\mu$ and asks what four-vector the electromagnetic field puts on the right — the answer is $qF^{\mu\nu}u_\nu$, and its time component is the power (2.5.46); that chapter also finds the missing momentum that Chapter 1.1 left unaccounted for, and it is in the field. Chapter 3.6 builds $T^{\mu\nu}$, the object that sources gravity, out of exactly the energy and momentum densities defined here — *energy* gravitates, not mass, which is why light bends. Chapter 4.1 reuses Worked Example 1 as evidence that photons carry momentum, and promotes $p^\mu = \hbar k^\mu$ from a structural suspicion to a law. Chapter 5.1 shows that $E = \gamma mc^2$ plus quantum mechanics makes particle number unfixable — you can always find $2mc^2$ somewhere — which is why fields replace particles. Chapter 5.7 computes a real cross-section in the Mandelstam variables §8.3 introduced. And Chapter 6.5 explains the 99% of the proton's mass that §9.2 could locate but not account for.